Contents

Chapter 1 Projective and Injective Modules, Free Modules

1.1 Free Modules

Definition 1.1.0.1

Universal Properties

♠ **Proposition 1.1.0.2** Suppose R be a ring, M be R-Mod and $I \in \mathcal{S}et$ be index, then we have a correspondence

$$Hom_{Mod-R}(R^{\oplus I}, M) \simeq Hom_{Set}(I, M)$$

◦ **Proof** ... □

1.2 Projective Modules

Definition 1.2.0.1 An object P of a category C is a **projective object** if it has the **left lifting property** against epimorphisms.

This means, that P is projective if for any morphism $f : P \to B$ and any epimorphism $q : A \to B$, f factors through

$$\begin{tikzcd} & A \arrow[d, "\forall q \in Epi"] \\ P \arrow[ur, dashed, "\exists"] \arrow[r, "\forall f"'] & B \end{tikzcd}$$

An equivalent way to say to say this is that

Definition 1.2.0.2 An object P is projective precisely if the $hom-functor$ $Hom(P, -)$ preserves epimorphisms.

With property of epimorphism, $P \to A$ is unique up to isomorphism.

♠ **Proposition 1.2.0.3** The followings are equivalent in $\mathcal{S}et$:

- Every object is projective;
- The axiom of choice holds.

◦ **Proof** ⇒ In ZF, AC ⇔ Every surjection $f : A \xrightarrow{B}$ has a section structure

$$g \circ f = id_A$$

We take $f = id_B, P = B$.

⇐ If AC holds, then we could give a function $P \to A$. Now we associate $f^{-1}(b)$ with $\bar{b} \in f^{-1}(b)$ by AC. Thus we could select $g : P \to \{\bar{b} | b \in B\}$ such that $f(x) = q(g(b))$.

$$
\begin{tikzcd}
 & C \arrow[d, "\subseteq"] \\
 & A \arrow[d, "q"] \\
P \arrow[r, "f"] \arrow[ur, dashed] \arrow[uur, dashed] & B
\end{tikzcd}
$$

□

Remark If we replace categorey of sets by another base topolos will behave essentially precisely like set, but in general will not validate the axiom of choice.

Homological algebra in such a more general context is abelian sheaves, and the theory is the theory of cohomology of abelian sheaf.

A main task is the resolution of R-mod.

Remark Although mostly, we have a forgetful functor $\mathcal{F} \to Sets$, but the functor is not full, so not preverse the functions.

Definition 1.2.0.4 The **projetive module** over R is the projective objects in $R - Mod$.

♠ **Proposition 1.2.0.5** The following properties are equivalent:

- P is projective module;
- If $g : B \to C$ is surjective, then
$$g_* : Hom(P, B) \to Hom(P, C)$$
is also surjective;
- For short exact sequence $0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$ Then,
$$0 \longrightarrow Hom_R(P, A) \longrightarrow Hom(P, B) \longrightarrow Hom(P, C) \longrightarrow 0$$
is exact;
- Short exact sequence $0 \to P \to F \to V \to 0$ is split exact;
- P is a direct item of free modules, i.e. there exist some V such that
$$F = P \oplus V$$
where F is free.

∘ **Proof**

- (2)⇔ (3)⇔ (4) By the discussion of exact sequence
- (3)⇔ (4) By the discussion of exact sequence.
- (1)⇔ (4) Consider the diagram

$$
\begin{tikzcd}
 & & A \arrow[d, two heads] \\
F = P \oplus V \arrow[urr, dashed, "Epi \Rightarrow \exists p"] \arrow[rr, dashed, bend right] & P \arrow[l, hook', "\iota"] \arrow[r] \arrow[ur, "p|P"] & B
\end{tikzcd}
$$

□

♠ **Proposition 1.2.0.6** Suppose P be a finite projective R-module. Then, there exists finite free module F, and finite module Q such that

$$F = P \oplus V$$

∘ **Proof** $\Leftarrow$
$\Rightarrow$ Consider the generators of P. □

Flatness of Projective Module

♠ **Proposition 1.2.0.7** Every projective module is flat.

∘ **Proof** Conisder the alternative definition □

Free Modules are Projective

♠ **Proposition 1.2.0.8** Let R be a ring.Every free R-module is projective.

∘ **Proof** Consider diagram

...

□

1.2.1 Sum

◇ **Lemma 1.2.1.1** ...

∘ **Proof** ... □

♠ **Proposition 1.2.1.2** (Criterion of Projective Property for Sum) Suppose $P = \sum P_i$, then

$$P \text{ is projective} \Leftrightarrow P_i \text{ is projective.}$$

∘ **Proof** ... □

★ **Corollary 1.2.1.3** The direct sum of projective modules is projective.

∘ **Proof** Omitted. □

1.2.2 Tensor

♠ **Proposition 1.2.2.1**

- Suppose P be a projective right R-module, then: $P \otimes_R - : {}_R M \to Ab$ is an exact functor.
- Similarly for M_R.

∘ **Proof** ... □

Flatness

Definition 1.2.2.2 Let R be a ring:

- (Flat) R-module is **flat** if $N_1 \to N_2 \to N_3$ is exact$\Rightarrow$ $M \otimes_R N_1 \to M \otimes_R N_2 \to M \otimes_R N_3$ is exact;
- (Faithfully Flat) Instead of $\Rightarrow$ by $\Leftrightarrow$;
- $R \to S$ is **flat/faithfully flat** if S is flat/faithfully flat module over R;

◇ **Lemma 1.2.2.3** Let R be a ring, $I, J \subseteq R$ be ideals. Let M be a flat R-module. Then

$$IM \cap JM = (I \cap J)M$$

∘ **Proof** Consider the exact sequence

$$0 \to I \cap J \to R \to R/I \oplus R/J \qquad \text{Iso III}$$

Tensoring with the flat module M, we obtain an exact sequence

$$0 \to (I \cap J) \otimes_R M \to M \to M/IM \oplus M/JM$$

We have

$$(I \cap J)M = Ker(M \to M/IM \oplus M/JM) = IM \cap JM$$

By the definition of morphism. □

◇ **Lemma 1.2.2.4** Let R be a ring, let $\{M_i, \varphi_{ii'}\}$ be a directed system of flat R-modules. Then

$$colim_i M_i$$

is a flat R-module.

∘ **Proof** Firstly: $\otimes$ commutes with colimits;
secondly: directed colimits are exact.
Thus

$$0 \to A \to B \to C \to 0 \Rightarrow 0 \to colim_i M_i \otimes A \to colim_i M_i \otimes B \to colim_i M_i \otimes C \to 0$$

□

◇ **Lemma 1.2.2.5** The composition of (faithfully) flat ring maps is (faithfully) flat.
If $R \to R'$ is (faithfully) flat, and M' is a (faithfully) flat R'-module, then M' is a (faithfully) flat R-module.

∘ **Proof**

- Consider

$$N_1 \to N_2 \to N_3$$

an exact complex of R-module. Then

$$R' \otimes_R N_1 \to R' \otimes_R N_2 \to R' \otimes_R N_3$$

Furthermore

$$M' \otimes_{R'} (R' \otimes_R N_1) \to M' \otimes_{R'} (R' \otimes_R N_2) \to M' \otimes_{R'} (R' \otimes_r N_3)$$

And we have

$$M' \otimes_{R'} (R' \otimes_R N) \cong (M' \otimes_{R'} R') \otimes_R N = M' \otimes_R N$$

- Omitted.

□

◇ **Lemma 1.2.2.6** Let M be an R-module. The following are equivalent:

- M is flat over R;
- For every injection of R-modules $N \subseteq N'$ the map

$$N \otimes_R M \to N' \otimes_R M$$

is injective.

- for every ideal $I \subseteq R$, the map

$$I \otimes_R M \to R \otimes_R M = M$$

is injective;

- For every finitely generated ideal $I \subseteq R$, the map

$$I \otimes_R M \to R \otimes_R M = M$$

is injective.

∘ **Proof**

- (1)⇒ (2) By definition;
- (2)⇒ (3) We take I, R as R-module.
- (3)⇒ (4) Trivial;
- (4)⇒ (1)

□

Directed Limit

◇ **Lemma 1.2.2.7** Let $\{R_i, \varphi_{ii'}\}$ be a system of rings over the directed set I. Let

$$R = colim_i R_i$$

- If M is an R-module such that M is **flat** as an R_i-module for all i. Then, M is flat as an R-module;
- For $i \in I$, let M_i be a flat R_i-moudle, and $i' \geq i$ and let $f_{ii'} : M_i \to M_{i'}$ be a $\varphi_{ii'}$-linear map such that

  $$f_{i'i''} \circ f_{ii'} = f_{ii''}$$

  Then, $M = colim_{i \in I} M_i$ is a flat R-module.

∘ **Proof** ... □

◇ **Lemma 1.2.2.8** Suppose that M is (faithfully) flat over R, and $R \to R'$ is a ring map. Then

$$M \otimes_R R'$$

is (faithfully) flat over R'.

∘ **Proof** ... □

◇ **Lemma 1.2.2.9** Suppose that $R \to R'$ be a faithfuly flat ring map, M be a module over R. Now, we set

$$M' = S' \otimes_S M$$

-
-

∘ **Proof** ... □

Via (faithfully) Flat over R'

◇ **Lemma 1.2.2.10** Suppose that M is (faithfully) flat over R, $R \to R'$ is a ring map. Then

$$M \otimes_R R'$$

is **(faithfully) flat over R'**.

∘ **Proof** ... □

◇ **Lemma 1.2.2.11** Let $R \to R'$ be a faithfully flat ring map. Let M be flat over R iff M' is flat over R.

∘ **Proof**

□

◇ **Lemma 1.2.2.12** Let R be a ring, $S \to S'$ be a flat map of R-algebras. Let M be a module oveer S. Set

$$M' = S' \otimes_S M$$

-
-

∘ **Proof** ... □

◇ **Lemma 1.2.2.13** ...

∘ **Proof** ... □

Equational Criterion of Flatness

◇ **Lemma 1.2.2.14** A module M over R is **flat** iff every relation in M is trivial.

∘ **Proof** ... □

Flatness of Items in Exact Sequence

◇ **Lemma 1.2.2.15** ...

∘ **Proof** ... □

Localization

Directed System of Faitfully Flat Algebras

♠ **Proposition 1.2.2.16** Let R be a ring, let $\{S_i, \varphi_{ii'}\}$ be a directed system of faifully flat R-algebras. Then

$$S = colim_i S_i$$

is a faithfully flat R-algebra.

∘ **Proof**

□

1.2.3 Projective Basis Theorem

▶ ***Theorem*** Let R be a ring, P is an R-module. The followings are equivalent:

- P is finitely generated projective module;
- Existence of projective basis: There exists a family of elements $\{x_i\}_{i=1}^n \subseteq P$, with a set of homomorphsims $\{\phi_i\}_{i=1}^n Hom_R(P, R)$, where ϕ_i is linear! and for all $x \in P$,

$$x = \sum \phi_i(x) x_i$$

∘ **Proof** ⇒ Suppose P is projecive, then $P \oplus Q = (R)^n$. Then, there exists:

$$i : P \to R^n \qquad p : R^n \to P$$

such that $p \circ i = id_P$. Then, $x = p(i(x)) = \sum a_j(x)p(e_j)$.
⇐ Suppose there exists such basis, then define

$$\begin{gathered} \psi : P \to R^n \qquad x \mapsto (\phi_1(x), \dots, \phi_n(x)) \\ \theta : R^n \to P \qquad (a_1, \dots, a_n) \mapsto \sum a_i x_i \\ \Rightarrow (\theta \circ \psi)(x) = x \end{gathered}$$

□

Note: Flatness
Note: Serre-Swan

1.2.4 Projective Resolution

Definition 1.2.4.1 Let M be R-module, then a **left resolution** is a complex $P_i = 0, i < 0$ with a map $\epsilon : P_0 \to M$ and the complex

$$\cdots \to P_1 \to P_0 \to M \to 0$$

is exact. An resolution is **projective** iff P_i is projective.

♠ **Proposition 1.2.4.2** Every R-module M has a projective resolution.
More generally, if an abelian category $\mathcal{A}$ has enough projectives, then every object M in $\mathcal{A}$ has a projective resolution.

∘ **Proof** Consider the following constructions: By selecting $M_{i-1} := ker(P_i \to P_{i-1})$ with new projective resolution $P_{i+1} \to M_i$. Now consider the compose

$$P_{i+1} \to M_i \to P_i$$

$$\begin{tikzcd} P_1 \arrow[rr] \arrow[dr] & & P_0 \arrow[r] & M \arrow[r] & 0 \\ & M_0 \arrow[ur] \arrow[dr] & & & \\ 0 \arrow[ur] & & 0 & & \end{tikzcd}$$

And P_{n+2} is projective. Thus

□

Comparision Theorem/ Homological Algebra's Fundamental Theorem

▶ ***Theorem*** Let $P \xrightarrow{\epsilon} M$ be a projective resolution of M and $f' : M \to N$ be a map in $\mathcal{A}$. Then, for every resolution $Q \xrightarrow{\eta} N$ there is a chain map $f : P \to Q$ lifting f' in the sense of

$$\eta \circ f_0 = f' \circ \epsilon$$

$$\begin{tikzcd} \dots \arrow[r] & P_1 \arrow[r] \arrow[d, dashed] & P_0 \arrow[r, "\epsilon"] \arrow[d, dashed, "f_0"] & M \arrow[r] \arrow[d, "f'"] & 0 \\ \dots \arrow[r] & Q_1 \arrow[r] & Q_0 \arrow[r, "\eta"] & N \arrow[r] & 0 \end{tikzcd}$$

Moreover, such f is unique up to **chain homotopy**.

- **Proof** The prove the existence of such chain P by induction:
  Basis Step: Consider the following square

$$\begin{tikzcd} P_0 \arrow[r, "\epsilon"] \arrow[d, dashed, "f_0"] & M \arrow[d, "f'"] \\ Q_0 \arrow[r, "\eta"] & N \end{tikzcd}$$

ϵ and η are epi. So the triangle

$$\begin{tikzcd} & Q_0 \arrow[d, "\eta"] \\ P_0 \arrow[ur, dashed] \arrow[r, "f\circ\epsilon"] & N \end{tikzcd}$$

Induction Step: Consider

$$\begin{tikzcd} P_{n+2} \arrow[r] \arrow[d, dashed, "f_{n+2}"] & P_{n+1} \arrow[r] \arrow[d, "f_{n+1}"] & P_n \arrow[d, "f_n"] \\ Q_{n+2} \arrow[r] & Q_{n+1} \arrow[r] & Q_n \end{tikzcd}$$

We use diagram chasing: Consider $p \in P_{n+2}$, given a f_{n+2} such that

$$d_{n+2} \circ f_{n+2} = f_{n+1} \circ d_{n+1}$$

Notice that $Im(f_{n+1} \circ d_{n+2}) \subseteq Ker(d_{n+1}) = Im(d_{n+2})$. Thus we have the following triangle

$$\begin{tikzcd} & Q_{n+2} \arrow[d] \\ P_{n+2} \arrow[ur, dashed, "f_{n+2}"] \arrow[r] & Im(d_{n+2}) \end{tikzcd}$$

□

Horseshow Lemma

▶ ***Theorem*** Suppose given a commutative diagram

$$\begin{tikzcd} & & & 0 \arrow[d] & \\ \dots P'_2 \arrow[r, "P_2"] & P'_1 \arrow[r] & P'_0 \arrow[r] & A' \arrow[r] \arrow[d] & 0 \\ & & & A \arrow[d] & \\ \dots P'_2 \arrow[r] & P'_1 \arrow[r] & P'_0 \arrow[r] & A' \arrow[r] \arrow[d] & 0 \\ & & & 0 & \end{tikzcd}$$

where the column is exact and the row are projective resolutions. Then, set

$$P_n := P'_n \oplus P''_n$$

The right hands column lift an exact sequence of complexes

$$0 \to P' \xrightarrow{i} P \xrightarrow{\pi} P'' \to 0$$

∘ **Proof** ... □

1.2.5 Projective Module over PID

♠ **Proposition 1.2.5.1** If a ring R is a PID,(especially, $R = \mathbb{Z}$) then, every projecive R-module is free.

∘ **Proof** Notice that: Every submodule of module over PID is free. Then, by the equivalent definition of projective module. □

1.2.6 Kaplansky's Theorem

◇ **Lemma 1.2.6.1** Suppose P be a projective module, $A \subseteq P$ be a submodule. If A is finitely/countably generated, then there exists an item Q such that:

-
-
-

∘ **Proof** ... □

▶ ***Theorem*** (Kaplansky) Every projective module is the direct sum of the finitely generated projective modules.

∘ **Proof** Firstly, consider the chain of submodules:

$$P_0 \subseteq P_1 \subseteq \cdots \subseteq P_i \subseteq \dots$$

where P_{i+1}/P_i is a □

▶ ***Theorem*** Let R be a ring. Then the following are equivalent:

- R is a local ring;
- Every projective module over R is free, and has **indecomposable decomposition**

$$M = \oplus M_i$$

such that for each maximal direct summand L of M, there exists a decomposition

...

1.3 Locally Free/Finite Locally Free Modules

Definition 1.3.0.1 Let R be a ring and M be an R-module:

- We say that M is **locally free** if we can cover $Spec(R)$ by **standard opens** $D(f_i), i \in I$ such that: M_{f_i} is a free R_{f_i}-module for all $i \in I$. i.e.

$$M_{f_i} = R_{f_i}^{\oplus n_i}$$

- We say that M is **finite locally free** if we can choose the covering such that M_{f_i} is finite free.
- M is **finite locally free** of **rank** r if

$$M_{f_i} \cong R_{f_i}^{r_i}$$

◇ **Lemma 1.3.0.2** Let R be a ring and let M be an R-module. The following are equivalent:

1. M is finitely presented, and R-flat;
2. M is finite projective;
3. M is a direct summand of a finite free R-module;
4. M is finitely presented and for all $p \in Spec(R)$, the localization M_p is free;
5. M is finite and locally free;
6. M is finitely presented and for all maximal ideals, $m \subseteq R$ the localization M_m is free;
7. M is finite, for every prime p, the module M_p is free, and the function

$$\rho_M : Spec(R) \to \mathbb{Z} \qquad p \mapsto dim_{\kappa(p)} M \otimes_R \kappa(p)$$

is **locally constant** in Zaraski topology.

∘ **Proof** We use this way to prove:

$$\begin{tikzcd}
(2) \arrow[r, Leftrightarrow] & (3) \arrow[r, Rightarrow] & (1) \arrow[r, Rightarrow] & (7) \arrow[r, Leftrightarrow] \arrow[d, Rightarrow] \arrow[dr, Rightarrow] & (6) \\
 & & (5) \arrow[u, Rightarrow] & (4) \arrow[l, Rightarrow] & (8)
\end{tikzcd}$$

- (2) $\Leftrightarrow$ (3): Consider the equivalent definition of projective module: P is projective $\Leftrightarrow F = P \oplus V$, notice that, by the construction of F, such F is also finite.
- (3) $\Rightarrow$ (1):If $R^{\oplus n} = M \oplus N$, and n is finite. We have

$$M = R^{\oplus n}/N$$

which is finitely presented. And M is projective $\Rightarrow - \otimes M$ is flat.

- (1) $\Rightarrow$ (7): Pick a prime p; $x_1, \ldots, x_r \in M$ which map t a basis of $M \otimes_R \kappa(p)$.
- (6)$\Leftrightarrow$ (7)

- (7)$\Rightarrow$ (8)
- (7)$\Rightarrow$ (4)
- (4)$\Rightarrow$ (5)
-

□

Hint R-flat+finite$\not\Rightarrow$ Projective.

Counterexample: With $R = C^\infty(\mathbb{R})$, $M = R_m = R/I$ where $m = \{f \in R | f(0) = 0\}$ and $I = \{f \in R | \exists \epsilon, \epsilon > 0 : f(x) = 0, \forall x, |x| < \epsilon\}$.

♠ **Proposition 1.3.0.3** If R be a reduced ring, and let M be an R-module. Then, equivalent to:

-

∘ **Proof** ... □

♠ **Proposition 1.3.0.4** Suppose: R is a local ring, M is a finite flat R-module. Then M is finite free.

∘ **Proof** ... □

♠ **Proposition 1.3.0.5** Let $R \to S$ be a flat local homomorphism of local rings. Let M be a finite R-module. Then, M is finite projective over R iff

---

∘ **Proof** ... □

♠ **Proposition 1.3.0.6** For any R-module P the following statements are equivalent:

- P is finite locally free
-
-
-

1.3.1 Faithfully flat descent for projectivity of modules

Question Suppose $\phi : R \to S$ be a ring homomorphism, M is R-module. Could we state, that $M \otimes_R S$ is projective S-module$\Rightarrow$ M is projective R-module?

In general not, but

♠ **Proposition 1.3.1.1** If ϕ is **faithfully flat**,

$$(0 \to M' \to M \to M'' \to 0) \in Exact(R - Mod) \Leftrightarrow (0 \to M' \otimes S \to M \otimes S \to M'' \otimes S \to 0) \in Exact(R - Mod)$$

then the statement holds.

∘ **Proof** ... □

1.4 Injective Modules

Definition 1.4.0.1 The **injective object** in $\mathcal{A}$ satisfies the universal lifting property: Given an injection $f : A \to B$ and a map $\alpha : A \to I$ there exists at least one map $\beta : B \to I$ such that $\alpha = \beta \circ f$

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "\alpha"] & B \arrow[dl, "\beta"] \\ & I & \end{tikzcd}$$

We say $\mathcal{A}$ has enough injectives if for every object A in $\mathcal{A}$, there is an injection $A \to I$ with I injective.

♠ **Proposition 1.4.0.2** In category $Sets$, every nonempty object is injective.

∘ **Proof** ...

□

Product

◇ **Lemma 1.4.0.3** If I_α is a family of injectives, so is $\prod I_\alpha$.

∘ **Proof** ... □

1.4.1 Baer's Criterion

▶ ***Theorem*** (Baer) A right R-module $E \in Mod - R$ is injective iff for every right ideal J of R, every map $J \to E$ can be extended to a map

$$R \to E$$

★ **Corollary 1.4.1.1** ...

∘ **Proof** ... □

Abelian Groups

★ **Corollary 1.4.1.2** $\mathcal{A}b$ has enough injectives.

∘ **Proof** ... □

1.4.2 Embedding Theorem

1.4.3 Injective Resolution

Definition 1.4.3.1 ...

Comparison Theorem

Adjoint Functors

◇ **Lemma 1.4.3.2** For every right R-module $M<$ the natural map

$$\tau : Hom_{Ab}(M, A) \to Hom_{mod-R}(M, Hom_{Ab}(R, A))$$

is an isomorphism, where

∘ **Proof** ... □

Definition 1.4.3.3 A pair of functions $L : \mathcal{A} \to \mathcal{B}$ and $R : \mathcal{B} \to \mathcal{A}$ are adjoint if there is a natural bijection for all $A \in \mathcal{A}$ and $B \in \mathcal{B}$ such that

$$\tau : Hom_{\mathcal{B}}(L(A), B) \xrightarrow{\cong} Hom_{\mathcal{A}}(A, R(B))$$

Natural: For $f : A \to A'$ and $g : B \to B'$,

$$\begin{array}{ccc} Hom_{\mathcal{B}}(L(A'), B) & Hom_{\mathcal{B}}(L(A), B) & Hom_{\mathcal{B}}(L(A), B') \\ \\ Hom_{\mathcal{A}}(\mathcal{A}', R(B)) & Hom_{\mathcal{A}}(A, R(B)) & Hom_{\mathcal{A}}(A, R(B')) \end{array}$$

★ **Corollary 1.4.3.4** ...

∘ **Proof** ... □

♠ **Proposition 1.4.3.5**

∘ **Proof** ... □

★ **Corollary 1.4.3.6** ...

∘ **Proof** ... □

1.5 Projective and Injective Objects in Abelian Category

1.5.1 Projective Object

Definition 1.5.1.1 Object $X \in C$ is **projective** if it satisfies the left lifting property for **epimorphism**.

$$\begin{tikzcd} & I \arrow[d, two heads] \\ A \arrow[ur, dashed] \arrow[r] & B \end{tikzcd}$$

♠ **Proposition 1.5.1.2** Let $X \in \mathcal{A}$, $\mathcal{A}$ be an abelian category. The followings are equivalent:

- X is projective.

- $Hom(X, -)$ is exact.

∘ **Proof**

- $\Rightarrow$ Consider the exact sequence

$$0 \longrightarrow C_1 \xrightarrow{f} C_2 \xrightarrow{g} C_3 \longrightarrow 0$$

Then the morphisms induced by the morphisms above

$$0 \longrightarrow Hom(X, C_1) \xrightarrow{f^*} Hom(X, C_2) \xrightarrow{g^*} Hom(X, C_3) \longrightarrow 0$$

Injective easy to verify, and surjective given by the diagram

$$\begin{tikzcd} & C_2 \arrow[d, "g"] \\ X \arrow[r] \arrow[ur, dashed] & C_3 \end{tikzcd}$$

$Im(f) = Ker(g)$ from exactness of $Hom(X, -)$.

- $\Leftrightarrow$ If $Hom(X, -)$ is exact, now consider the following exact sequnece

$$0 \to Ker(g) \to A \to B \to 0$$

And apply $Hom(X, -)$, surjectivity of $Hom(X, A) \to Hom(X, B)$.

□

Note Without the projective property, $Hom(X, -)$ is just left-exact. See discussion here:

Projective Resolution

1.5.2 Injective Object

Definition 1.5.2.1 An object is **injective** if it satisfies the right lifting property for **monomorphism**

$$\begin{tikzcd} B \arrow[r] & A \\ P \arrow[u, dashed] \arrow[ur, dashed] & \end{tikzcd}$$

Preservation of Injective Objects

◇ **Lemma 1.5.2.2** Given a pair of additive adjoint functors

$$(L)$$

betweeen abelian categories such that the left adjoint L is a **left exact functor**, then the right adjoint preserve injective objects.

∘ **Proof** ... □

♠ **Proposition 1.5.2.3** ...

∘ **Proof** ... □

Existence of Enough Injectives

Given some categories with enough injectives:

-
-

◇ **Lemma 1.5.2.4** ...

- ∘ **Proof** ... □

Constructions

♠ **Proposition 1.5.2.5**

- The (small) product of injective objects is injective;
- The direct sum of finite injective objects is injective;

∘ **Proof**

- Consider the following diagram

$$\begin{tikzcd} & \prod I_i \arrow[d] \\ & I_i \arrow[d, two heads] \\ A \arrow[r] \arrow[ur] \arrow[uur, dashed] & B \end{tikzcd}$$

- $$\begin{tikzcd} & \prod_{i \in I} I_i \arrow[dd, bend left, two heads] \\ & \sum I_i \arrow[u] \arrow[d, two heads] \\ A \arrow[r] \arrow[ur] & B \end{tikzcd}$$

□

And the injectivity of arbitrary direct sum can be judged with Baer's criterion.

Injective Resolution

Definition 1.5.2.6 An **injective** resolution is a cochain complex $J^* \in Ch^*(\mathcal{A})$ equipped with a **quasi-isomorphism**

$$i : X \xrightarrow{\sim} J^*$$

such that $J^n \in \mathcal{A}$ is an injective object for all $n \in \mathbb{N}$. i.e.Quasi-isomorphism

$$\begin{tikzcd} X \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \dots \arrow[r] & 0 \arrow[r] \arrow[d] & \dots \\ J^0 \arrow[r] & J^1 \arrow[r] & \dots \arrow[r] & J^n \arrow[r] & \dots \end{tikzcd}$$

♠ **Proposition 1.5.2.7** If $\mathcal{A}$ has enough injectives, then every $X \in \mathcal{A}$ has an injective resolution.

∘ **Proof** We have □

♠ **Proposition 1.5.2.8** ...

∘ **Proof** ... □

Injective Resolution

Definition 1.5.2.9 For $X \in \mathcal{A}$ an object, an **injective resolution** is a cochain complex $J^* \in Ch^*(\mathcal{A})$ equipped with a quasi-isomorphism

$$i : X \xrightarrow{\sim} J^*$$

such that $J^n \in \mathcal{A}$ is an **injective object** for all $n \in \mathbb{N}$.

$$\begin{tikzcd} X \arrow[r] \arrow[d, "i^0"] & 0 \arrow[r] \arrow[d] & \dots \arrow[r] & 0 \arrow[r] \arrow[d] & \dots \\ J^0 \arrow[r, "d^0"] & J^1 \arrow[r, "d^1"] & \dots \arrow[r] & J^n \arrow[r, "d^n"] & \dots \end{tikzcd}$$

Projective Resolution

Definition 1.5.2.10 For $X \in \mathcal{A}$ is an object, a **projective resolution** of X is a chain ccomplex $J_* \in Ch_\bullet(\mathcal{A})$ equipped with a quasi-isomorphism

$$p : J_* \xrightarrow{\sim} X$$

such that $J_n \in \mathcal{A}$ is a projective object for all $n \in \mathbb{N}$.

$$\begin{tikzcd} \dots \arrow[r] \arrow[d] & J_n \arrow[r] \arrow[d] & \dots \arrow[r] & J_1 \arrow[r] \arrow[d] & J_0 \arrow[d] \\ \dots \arrow[r] & 0 \arrow[r] & \dots \arrow[r] & 0 \arrow[r] & X \end{tikzcd}$$

♠ **Proposition 1.5.2.11** Let $\mathcal{A}$ be an abelian category with enough injectives, such as $R-Mod$ for some ring R, then every object $X \in \mathcal{A}$ has an injective resolution.

∘ **Proof** ... □

♠ **Proposition 1.5.2.12** Let $\mathcal{A}$ be an abelian category with enough projectives(such as R-Mod for some ring R). Then every object $X \in \mathcal{A}$ has a projective resolution.

∘ **Proof** ... □

1.5.3 Resolutions, and Ext^1

Suppose $\mathcal{A}$ be an abelian category.

Left Resolution

Definition 1.5.3.1 A **left resolution** of $M \in \mathcal{A}$ consists of:

- Chain complex $A_\bullet$;
- Morphism $\epsilon : A_0 \to M$;
- Exactness

$$\cdots \to A_2 \to A_1 \to A_0 \xrightarrow{\epsilon} M \to 0$$

Free Resolution

Definition 1.5.3.2 Let $M \in \mathcal{A}$. A **free resolution**

Conclusions

♠ **Proposition 1.5.3.3** Every finitely generated abelian group admits a free resolution with length 2.

- **Proof** Take the free object generated by □

Some of contents will be given in:Syzygies.

1.5.4 Universally Injective Module Maps

Definition 1.5.4.1

- Let $f : M \to N$ be a map of R-modules.
  Then f is called **universally injective** if for every R-module Q, the map
  $$f \otimes_R id_Q : M \otimes_R Q \to N \otimes_R Q$$
  is injective.
- A sequence $0 \to M_1 \to M_2 \to M_3 \to 0$ of R-modules is called **universally exact** if it is exact, and $M_1 \to M_2$ is universally injective.

♣ **Example 1.5.4.2** ...

▶ ***Theorem*** Let

$$0 \to M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \to 0$$

be an exact sequence of R-modules. The following are equivalent:

- Sequence $0 \to M_1 \to M_2 \to M_3 \to 0$ is universally exact;
- For every finitely presented R-module Q, the sequence
  $$0 \to M_1 \otimes_R Q \to M_2 \otimes_R Q \to M_3 \otimes_R Q \to 0$$
- Given elements $x_i \in M_1; i = 1, \dots, n; y_i \in M_2, j = 1, \dots, m$. And $a_{ij} \in R(i = 1, \dots, n, j = 1, \dots, m)$ such

that for all i

$$f_1(x_i) = \sum_j a_{ij} y_j$$

there exists $z_i \in M_1$ such that for all i:

...

- Given a commutative diagram of R-module maps

...

-
-

∘ **Proof** ... □

◇ **Lemma 1.5.4.3** Let

$$0 \to M_1 \to M_2 \to M_3 \to 0$$

be an exact sequence of R-modules. Suppose that: M_3 is finite presentaion. Then,

...

∘ **Proof** ... □

1.6 Flatness

1.6.1 Exact Sequence

♠ **Proposition 1.6.1.1** (Exactness of Flatness) Suppose $0 \to A \to B \to C \to 0$ an exact sequence with A, C flat. Then, B is flat.

∘ **Proof** We just consider the following diagram: Notice that the tensor product $-\otimes X$ is in general right-exact, $X = A, B, C$ we just need to draw the items, for the short exact sequence $0 \to M' \to M$ we have

$$\begin{tikzcd} & M' \otimes A \arrow[r] \arrow[d, "f"] & M' \otimes B \arrow[r] \arrow[d, "g"] & M' \otimes C \arrow[r] \arrow[d, "h"] & 0 \\ 0 \arrow[r] & M \otimes A \arrow[r] & M \otimes B \arrow[r] & M \otimes C & \end{tikzcd}$$

We can get an exact sequence $ker(f) \to ker(g) \to ker(h)$, and $ker(f) = ker(h) = 0 \Rightarrow ker(g) = 0$.
Another solution is to use the statement with 9-lemma:

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & M' \otimes A & \longrightarrow & M' \otimes B & \longrightarrow & M' \otimes C & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & M \otimes A & \longrightarrow & M \otimes B & \longrightarrow & M \otimes C & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & M'' \otimes A & \longrightarrow & M'' \otimes B & \longrightarrow & M'' \otimes C & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

Column 1,3 are exact, so is column 2.

□

1.6.2 Characterizing Flatness

◇ **Lemma 1.6.2.1** Let M be an R-module. The following are equivalent:

- M is flat;
- If $f : R^n \to M$ is a module map, $x \in Ker(f)$, then there are module maps

$$h : R^n \to R^m \qquad g : R^m \to M$$

  such that $f = g \circ h$ and $x \in Ker(h)$.
-
-

∘ **Proof**

□

▶ ***Theorem*** (Lazard's Theorem) Let M be an R-module. Then,
M is flat⇔ It is the colimit of a directed system of free finite R-modules.

∘ **Proof** ... □

Moreover,

- https://arxiv.org/pdf/1109.0439 'The Lazard-Linke's Theorem on quasi-coherent sheaves.

1.6.3 Flatness over PID

Chapter 2 Chain Complexes

2.1 Some of Category Theory

2.1.1 Ab-Category

Definition 2.1.1.1 A category $\mathcal{A}$ is called an $\mathcal{A}b$-category if every *hom*-set $Hom_{\mathcal{A}}(A, B)$ in $\mathcal{A}$ is given the structure of an abelian group in such a way that composition distributes over addition.

In particular, given a diagram in $\mathcal{A}$ of the form

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, shift left, "g'"] \arrow[r, shift right, "g"'] & C \arrow[r, "h"] & D \end{tikzcd}$$

We have

$$h(g + g')f = hgf + hg'f$$

2.1.2 Preadditive Category

(Stack Project, Homological Algebra, 3)

Definition 2.1.2.1 A category $\mathcal{A}$ is **preadditive** if each morphism set $Mor_{\mathcal{A}}(x, y)$ is endowed with a structure of an abelian group such that:

- Each composition

$$Mor(x, y) \times Mor(y, z) \to Mor(x, z)$$

is bilinaer($(f + g)\circ h = f\circ h + g\circ h, f\circ(g + h) = f\circ g + f\circ h$)

Additive functor

Definition 2.1.2.2

- An addive functor $F : \mathcal{B} \to \mathcal{A}$ between $\mathcal{A}b$-categories $\mathcal{B}$ and $\mathcal{A}$ is a functor such that each

$$Hom_{\mathcal{B}}(B', B) \to Hom_{\mathcal{A}}(A', A)$$

is a group homomorphism;

◇ **Lemma 2.1.2.3** Let $\mathcal{A}$ be a preadditive. Let $x \in Ob(\mathcal{A})$, the following are equivalent:

- x is an initial object;
- x is a final object;
- $id_x = 0 \in Mor_{\mathcal{A}}(x, x)$

Furthermore, if such 0 exists, then: A morphism $\alpha : y \to z$ factors through $0 \Leftrightarrow \alpha = 0$

- **Proof** ... □

Such object is called **zero object** in preadditive category.

Remark Existence:

◇ **Lemma 2.1.2.4** (Isomorphism of Finite Product, Coproduct) In preadditive category

- $x \times y$ exists$\Leftrightarrow$ $x \sqcup y$ exists;
- we have

$$x \sqcup y \cong x \times y$$

∘ **Proof**

- If $z = x \sqcup y$, define $p := i \sqcup 0, p := i' \sqcup 0$
- If $z = x \times y$, define $i := id_x \times 0, i' = 0 \times id_y$

$$\begin{tikzcd}
x \arrow[rr, "id_x"] \arrow[dr, "i"] & & x \\
& z \arrow[ur, "p"] \arrow[dr, "p'"] & \\
y \arrow[ur, "i'"] \arrow[rr, "id_y"] & & y
\end{tikzcd}$$

□

Definition 2.1.2.5 (Direct Sum) In preadditive category, such $x \oplus y := x \times y = x \sqcup y$.

◇ **Lemma 2.1.2.6**

∘ **Proof** ... □

2.1.3 Additive Funtors, Additive Category

Definition 2.1.3.1

- An **additive category** is an $\mathcal{A}b$-category $\mathcal{A}$ with:
  - Zero Object $0 \in \mathcal{A}$;
  - Product $A \times B$;

◇ **Lemma 2.1.3.2** For additive category, we have

$$A \times 0 \cong A, A \in \mathcal{A}$$

∘ **Proof** Consider

$$\begin{tikzcd}
& A & \\
A \times 0 \arrow[ur, "p_A"] \arrow[dr] & & A \arrow[ll, "f"] \arrow[ul, "id_A"'] \arrow[dl] \\
& 0 &
\end{tikzcd}$$

We have $p_A \circ f = id_A, f \circ p_A = id_{A\times 0}$ □

◇ **Lemma 2.1.3.3** (Product Map) If $f \in Mor_{\mathcal{A}}(B, C), g \in Mor_{\mathcal{A}}(D, E)$, then we have

$$f \times g : B \times D \to C \times E$$

∘ **Proof** Consider following diagram

$$\begin{tikzcd} B \arrow[r] & C \\ B \times D \arrow[u] \arrow[d] \arrow[r, dashed] & C \times E \arrow[u] \arrow[d] \\ D \arrow[r] & E \end{tikzcd}$$

□

♠ **Proposition 2.1.3.4** If $\mathcal{A}$ is an additive category, then the product exists, with

$$A \times B = A \sqcup B$$

∘ **Proof** Firstly, $A \times B$ has following mappings $\iota_A : A \cong A \times 0 \to A \times B$(B resp.); And then for $p_A : P \to A, p_B : P \to B$ we have

$$\iota_A p_A + \iota_B p_B : P \to A \times B$$

Thus, $A \times B$ is the coproduct. □

Also take the following notion

$$A \times B \cong A \oplus B$$

Then,

♠ **Proposition 2.1.3.5** The following statements are equivalent:

- Category C is additive;
- The structure maps
  - $$\begin{aligned} i_X &: X \to X \oplus Y \\ i_Y &: Y \to X \oplus Y \end{aligned}$$
  - $$\begin{aligned} p_X &: X \oplus Y \to X \\ p_Y &: X \oplus Y \to Y \end{aligned}$$

  with

–

$$p_X \circ i_X = 1_X \qquad p_Y \circ i_Y = 1_Y$$
$$p_X \circ i_Y = 0 \qquad p_Y \circ i_X = 0$$

– Completeness

$$1_{X \oplus Y} = i_X \circ p_X + i_Y \circ p_Y$$

∘ **Proof** Omitted. □

In additive category, the morphism between direct products can be represented as matrix

$$f : A \oplus B \to C \oplus D$$
$$f = \begin{pmatrix} p_W \circ f \circ i_X & p_W \circ f \circ i_Y \\ p_Z \circ f \circ i_X & p_Z \circ f \circ i_Y \end{pmatrix} = \begin{pmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{pmatrix}$$

2.1.4 Karoubian Categories

(Stack Project, HA, 4)

Definition 2.1.4.1 Let C be a preadditive category.
We say C is **Karoubian** if every idempotent endomorphism ($f^2 = f$) of an object of C has a kernel.

$$f^2 = f \Rightarrow \exists ker(f), (0 \to ker(f) \to x \to y)$$

♠ **Proposition 2.1.4.2** (Equivalent Conditions for Karoubian Categories) The followings are equivalent:

- C is Karobian;
- Each idempotent endomorphism of an object C has a cokernel;
- Given an idempotent endomorphism $p : z \to z$, there exists a direct sum decomposition $z = x \oplus y$ such that: p corresponds to the projecion onto y

---

∘ **Proof**

- (1) ⇒ (2)
- (2) ⇒ (3)
- (3) ⇒ (1)

□

♠ **Proposition 2.1.4.3** (Criterias for Karoubian Category) Let D

- If D has countable products and kernels of maps which have a right inverse, then Karoubian; D
- If D has countable ciproduct and cokernels of maps which have a left inverse, then Karoubian; D

∘ **Proof**

-
-

□

Warning Preadditive Category ⊂ Karoubian Category⊂ Additive Category

♣ **Example 2.1.4.4** Category of free modules.

♣ **Example 2.1.4.5** Category $\mathcal{S}et$.

2.1.5 Abelian Category

Definition 2.1.5.1 Define: For $f \in Mor_{\mathcal{A}}(B, C)$,

- (Kernel) The **kernel** is the universal object given by $i : Ker(f) \to B$ with $fi = 0$
- (Cokernel) The **cokernel** is the universal object given by $j : C \to Coker(f)$ with $jf = 0$.

And:

- f is mono iff $fg = fh \Rightarrow g = h$
- f is epi iff $gf = hf \Rightarrow g = h$

Then,

Definition 2.1.5.2 The **abelian category** satisfies:

- f is Mono/injective⇔ $Ker(f) = 0$;
- f is Epi/surjecitve⇔ $Coker(f) = 0$;

Define:

$$Coim(f) := Coker(Ker(f) \to B)$$
$$Im(f) := Ker(C \to Coker(f))$$

Remark Sometimes take the notion:

- inj.:$Ker(f) = 0$;
- sur.:$Coker(f) = 0$;

♠ **Proposition 2.1.5.3** Category $\mathcal{A}$ is abelian iff it is additive, all kernels and cokernels exist, and the natural map

$$Coim(f) \to Im(f)$$

is an isomorphism for all morphisms f of $\mathcal{A}$.

∘ **Proof** Notice: $Coim(f) = Coker(Ker(f) \to B) = Ker(C \to Coker(f)) = Im(f)$, now consider following property

$$Ker(f) \to B \to Coim(f) = Im(f) \to C \to Coker(f)$$

Then, equivalent by universal property. Omitted. □

We could define the exactness:

Definition 2.1.5.4 For abelian category $\mathcal{A}$, we could define the **exactness**

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

With $Im(f) = Ker(g)$.

◇ **Lemma 2.1.5.5** Let: $\mathcal{A}$ be a preadditive category.
The addition on sets of morphisms make $\mathcal{A}^{opp}$ into a preadditive category.
Furthermore, $\mathcal{A}$ is preadditive/additive/abelian$\Leftrightarrow$ $\mathcal{A}^{op}$ is preadditive/additive/abelian ;

∘ **Proof** ... □

◇ **Lemma 2.1.5.6** (Finite Limimts, Colimits) The finite limits, colimits exist in the abelian category.

∘ **Proof**

- Limit
- Colimit

□

▶ ***Theorem*** ...

∘ **Proof** ... □

2.1.6 Pullback,Pushout Diagram

Definition 2.1.6.1 The **intersection/pullback** of objects $x, y \in \mathcal{A}$, which satisfies the following diagram

$$\begin{tikzcd}
x \cap y \arrow[r, "p_1"] \arrow[d, "p_2"] \arrow[dr, phantom, "\lrcorner", very near start] & x \arrow[d, hook] \\
y \arrow[r, hook] & z
\end{tikzcd}$$

And the **cointersection/pushout**

$$\begin{tikzcd}
z \arrow[r, two heads] \arrow[d, two heads] & x \arrow[d] \\
y \arrow[r] & x \cup_z y
\end{tikzcd}$$

2.1.7 Lemmas

3-Lemma

Snake Lemma

♠ **Proposition 2.1.7.1** ...

∘ **Proof** ... □

4-Lemma

5-Lemma

2.1.8 Grothendieck Category

`https://ncatlab.org/nlab/show/additive+and+abelian+categories`
`https://ncatlab.org/nlab/show/Grothendieck+category`

Definition 2.1.8.1 (Grothendieck)

- (AB1) Existence of kernel,cokernel;
- (AB2) Isomorphism $Coim(f) \to Im(f)$,where $Coim(f) = Coker(Ker(f) \to A), Im(f) = Ker(A \to Coker(f))$;

  (AB1)+(AB2)=Abelian
- (AB3) An abelian category with all coproducts; Hence all colimits;
- (AB4) AB3 Satisfied, coproduct of monomorphism is monic;
- (AB5) AB3 Satisfied, filtered limits of exact seuqneces exist and are exact;
- (AB6) AB3 Satisfied, such that for increasing directed set J, $(B^j)_{j\in J}$ and families $B^j = (B_i^j)_{i\in I_j}$(Thus, $B = ((B_i^j)_{i\in I_j})_{j\in J}$) then, we have

$$\cap_{j\in J}(\sum_{i\in I_j} B_j^i) = \sum_{(i_j)\in\prod I_j}(\cap_{j\in J} B_{i_j}^i)$$

- (AB3*) DF
- (AB4*) DF
- (AB5*) DF
- (AB6*) DF

2.1.9 Extension

Definition 2.1.9.1

- An **extension** of A by B in abelian category $\mathcal{A}$ is an (short) exact sequence

$$0 \to A \to B \to C \to 0$$

- The morphism between extensions: For short exact sequences $0 \to A \to E \to B \to 0; 0 \to A \to F \to B \to 0$. And the following diagram commutes:

$$\begin{array}{ccccc} 0 & A & E & B & 0 \\ \\ 0 & A & F & B & 0 \end{array}$$

- The isomorphic class:Let $A, B \in Ob(\mathcal{A})$, the set of isomorphic classes of extension of B by A denoted

$$Ext_{\mathcal{A}}(B, A)$$

  is called the **Ext-group**.

Actually, we turn $Ext(-,-)$ into functor

- Pullback
- Pushout

◇ **Lemma 2.1.9.2** The construction

$$(E_1, E_2) \mapsto E_1 + E_2$$

defines a commutative group law on $Ext_{\mathcal{A}}(B, A)$, which is functorial on each variable.

- ∘ **Proof** ... □

♠ **Proposition 2.1.9.3** (Six-Item Exact Sequence) There exists long exact sequence

$$\begin{array}{ccccc} 0 & Hom(C, X) & Hom(B, X) & Hom(A, X) & \\ & Ext(C, X) & Ext(B, X) & Ext(A, X) & 0 \end{array}$$

∘ **Proof** ... □

♠ **Proposition 2.1.9.4** (Six-Item Exact Seuquence)

$$\begin{array}{ccccc} 0 & Hom(X, A) & Hom(X, B) & Hom(X, C) & \\ & Ext(X, A) & Ext(X, B) & Ext(X, C) & 0 \end{array}$$

∘ **Proof** ... □

2.1.10 Additive Functors

◇ **Lemma 2.1.10.1** Let $\mathcal{A}, \mathcal{B}$ abelian categories, $F : \mathcal{A} \to \mathcal{B}$ is functor. Then, the followings are equivalent

-
-
-

∘ **Proof** ... □

◇ **Lemma 2.1.10.2** ...

- ∘ **Proof** ... □

2.1.11 Localization

2.1.12 Jordan-Hölder

2.1.13 Serre Subcategories

Definition 2.1.13.1 Let $\mathcal{A}$ be an abelian category:

- A **Serre subcategory** of $\mathcal{A}$ is a nonempty full subcategory C such that for every exact sequence

$$A \longrightarrow B \longrightarrow C$$

with $A.C \in Ob(\mathcal{A})$ such that

$$A, C \in Ob(C) \Rightarrow B \in Ob(C)$$

- A **weak Serre subcategory** $\mathcal{A}$ is a nonempty full subcategory C of $\mathcal{A}$ such that: Given any exact sequence

$$A_0 \to A_1 \to A_2 \to A_3 \to A_4$$

with $A_0, A_1, A_3, A_4 \in Ob(C) \Rightarrow A_2 \in Ob(C)$.

◇ **Lemma 2.1.13.2** (Equivalent Condition of Serre Subcategory) Let $\mathcal{A}$ be an abelian category, C be a subcategory of C. Then, the following condition are equivalent: C is Serre subcategory$\Leftrightarrow$

- $0 \in Ob(C)$;
- C is a strictly full subcategory of $\mathcal{A}$;
- Any subobject or quotient of C is an object of C;
- If $A \in Ob(\mathcal{A})$ is an extension of objects of C, then $A \in Ob(C)$.

∘ **Proof** $\Leftarrow$
$\Rightarrow$ Omitted. □

◇ **Lemma 2.1.13.3** (Equivalent Condition of Weak Serre Subcategories)

-
-
-
-

∘ **Proof** ... □

Serre Subcategory by Morphism

◇ **Lemma 2.1.13.4** Let $\mathcal{A}, \mathcal{B}$ be abelian categories, and $F : \mathcal{A} \to \mathcal{B}$ be an exact functor.
Then, the **full subcategory** specified by $F(C) = 0$ forms a Serre subcategory of A.

∘ **Proof** Notice that the functor is exact. □

Definition 2.1.13.5 Let $\mathcal{A}, \mathcal{B}$ be an exact functors, then the full subcategory of objects C such that $F(C) = 0$ is called the **kernel** of the functor F. Sometimes denoted by

$$ker(\mathcal{F})$$

Quotient

◇ **Lemma 2.1.13.6** (Categoricial Quotient) Let $\mathcal{A}$ be an abelian category, $\mathcal{C} \subseteq \mathcal{A}$ be a Serre subcategory, there exists an abelian category and an exact functor

$$F : \mathcal{A} \to \mathcal{A}/\mathcal{C}$$

which is **essentially surjective** and whose kernel is $\mathcal{C}$.

The category $\mathcal{A}/\mathcal{C}$ is specified by the following universal property: For any exact functor

$$G : \mathcal{A} \to \mathcal{B}$$

Such that $\mathcal{C} \subseteq Ker(\mathcal{G})$, there exists a factorization $G = H \circ F$ for a unique exact functor $H : \mathcal{A}/\mathcal{C} \to \mathcal{B}$.

∘ **Proof** ... □

◇ **Lemma 2.1.13.7** Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $F : \mathcal{A} \to \mathcal{B}$ be an exact functor, let $\mathcal{C} \subseteq \mathcal{A}$ be Serre subcategory contained in the kernel of F. Then,

$$\mathcal{C} = Ker(F) \Leftrightarrow \bar{F} : \mathcal{A}/\mathcal{C} \to \mathcal{B} \text{ is faithful}$$

∘ **Proof** ... □

2.1.14 K-Groups of Abelian Category

0th Abelian Category

Definition 2.1.14.1 Let $\mathcal{A}$ be an abelian category, then, denote $K_0(\mathcal{A})$ the **zeroth K-group** of $\mathcal{A}$ and for exact sequence

Remark Another way, we to say this is that: There exists a presentation

$$\oplus_{(A\to B\to C)\in Ses} Z[A \to B \to C] \to \oplus_{A\in Ob(\mathcal{A})} Z[A] \to K_0(A) \to 0$$

◇ **Lemma 2.1.14.2**

∘ **Proof** ... □

◇ **Lemma 2.1.14.3** ...

∘ **Proof** ... □

◇ **Lemma 2.1.14.4** ...

∘ **Proof** ... □

2.1.15 Cohomological δ-Functor

Definition 2.1.15.1 Let $\mathcal{A}, \mathcal{B}$ be abelian categories, a **cohomological δ-functor** or simly **δ-functor** from $\mathcal{A} \to \mathcal{B}$ is given by the following datas:

- A collection
$$F^n : \mathcal{A} \to \mathcal{B}, n \geq 0$$
be additive functors, and
- For any short exact sequence $0 \to A \to B \to C \to 0$ of $\mathcal{A}$, a collection
$$\delta_{A \to B \to C} : F^n(C) \to F^{n+1}(A), n \geq 0$$
of morphisms of $\mathcal{B}$.

These data are assumed to satisfy the following axioms:

- For any short exact sequence as above, we have the **long exact sequence**
$$\begin{tikzcd} 0 \arrow[r] & F^0(A) \arrow[r] & F^0(B) \arrow[r] & F^0(C) \arrow[dll] \\ & F^1(A) \arrow[r] & F^1(B) \arrow[r] & F^1(C) \arrow[dll] \\ & \dots & & \end{tikzcd}$$
- For every morphism of short exact sequence $(0 \to A \to B \to C \to 0) \to (0 \to A' \to B' \to C' \to 0)$ of short exact sequences, the diagram
$$\begin{tikzcd} F^n(C) \arrow[r, "\delta_{A \to B \to C}"] \arrow[d] & F^{n+1}(A) \arrow[d] \\ F^n(C') \arrow[r, "\delta_{A' \to B' \to C'}"] & F^{n+1}(A') \end{tikzcd}$$

This implies, that $F^0(-)$ is left-exact.

Morphism of δ-Functors

Definition 2.1.15.2 ...

Universal δ-Functor

Definition 2.1.15.3 ...

◇ **Lemma 2.1.15.4** ...

∘ **Proof** ... □

◇ **Lemma 2.1.15.5** ...

∘ **Proof** ... □

2.2 Chain Complex

Definition 2.2.0.1

- A $\mathbb{Z}$-**graded chain complex** in $R-Mod$ is:
  - A collection of obejcts $\{C_n\}_{n\in\mathbb{Z}}$;
  - Morphisms $\partial_n : C_n \to C_{n-1}$ such that

$$\partial_n \circ \partial_{n+1} = 0, n \in \mathbb{N}$$

  Then:
  * ∂_n are called the **differentials** or **boundary maps**;
  * The elements of C_n are called n-chains;
  * For $n \geq 1$ the elements in the kernel

$$Z_n := Z_n(C_\bullet) = Ker(\partial_n)$$

    are called **n-cycles**;
  * The image

$$B_n := B_n(C_\bullet) = im(\partial_{n+1})$$

    of $\partial_n : C_n \to C_{n-1}$ are called the **n-boundaries**;

  There exists an inclusion $\partial^2 = 0 \Rightarrow 0 \hookrightarrow B_n \hookrightarrow Z_n \hookrightarrow C_n$; Define $H_n := Z_n/B_n$, then the **chain homology** of C_*

$$0 \to B_n \to Z_n \to H_n \to 0$$

Then,

Definition 2.2.0.2

- A **chain map** $f : V_* \to W_*$ is a collection of morphism $(f_n : V_n \to W_n)_{n\in\mathbb{Z}}$ in $\mathcal{A}$ such that all the diagram

$$\begin{tikzcd} V_{n+1} \arrow[r] \arrow[d] & V_n \arrow[d] \\ W_{n+1} \arrow[r] & W_n \end{tikzcd}$$

- Take the notation of $Ch(R-Mod)$ for the category. Furthermore, for the category $\mathcal{A}$, take the notion of $Ch(\mathcal{A})$.

Especially, when $R = \mathbb{Z}$, is the chain complex of abelian groups.

♠ **Proposition 2.2.0.3** $Ch(\mathcal{A})$ is an abelian category, iff $\mathcal{A}$ is an abelian category.

∘ **Proof** $\Rightarrow$ Omitted. There exists a faithful funcotor $\mathcal{A} \to Ch(\mathcal{A})$;
$\Leftarrow$ $\mathcal{A}$ is additive, with structure induced by $\mathcal{A}$. And the biproduct is well-defined. The kernel commutes with □

★ **Corollary 2.2.0.4** $Ch(R-Mod)$ is an abelian category.

∘ **Proof** Omitted. □

Exactness

◇ **Lemma 2.2.0.5** $Ker, Coker$ commutes with component operation $(-)_n : Ch(\mathcal{A}) \to \mathcal{A}$.

- ∘ **Proof**
- Ker:
- $Coker$:

□

♠ **Proposition 2.2.0.6** The followings are equivalent:

- $0 \to A \to B \to C \to 0$ is exact in $Ch(\mathcal{A})$;
- $0 \to A_n \to B_n \to C_n \to 0$ is exact for each $n \in \mathbb{Z}$ in $\mathcal{A}$;

∘ **Proof** By definition, omitted. □

♠ **Proposition 2.2.0.7** Suppose $A, B, C \in \mathcal{A}$, $\mathcal{A}$ abelian, $0 \to A \to B \to C \to 0$ is exact, then

$$0 \to A \to B \to C \to 0 \text{ splits} \Leftrightarrow B = A \oplus C$$

∘ **Proof** Consider the following diagram

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r, "f"] \arrow[dr, shift left] & B \arrow[r, shift right, "g"'] \arrow[d, shift right] & C \arrow[r] \arrow[l, dashed, shift right] \arrow[dl, shift left] & 0 \\
& & A \oplus C \arrow[ul, shift left] \arrow[u, dashed, shift right] \arrow[ur, shift left] & &
\end{tikzcd}$$

Take $f\pi_A + h\pi_B : A \oplus C \to B$

π_A, π_B satisfied by 2.1.3. □

♠ **Proposition 2.2.0.8** (Left-Exactness of $Hom(-, X)$) Suppose $X \in \mathcal{A}$, then $Hom(-, X)$ is left-exact: For exact

$$0 \to A \to B \to C \to 0$$

We get the left-exact sequence

$$0 \to Hom(C, X) \to Hom(B, X) \to Hom(A, X)$$

- ∘ **Proof** The morphisms induced by the morphims above.
- $Hom(C, X) \to Hom(B, X)$ is injective: We have

$$h \circ g = 0, g \text{ is surjective} \Rightarrow h = 0$$

- $Im(f^*) = Ker(g^*)$:
  - $Im(g^*) \subseteq Ker(f^*)$, easy to verify, that for $h \in Hom(C, X)$,

$$f^* g^*(h) = h \circ g \circ f = 0$$

– $Ker(f^*) \subseteq Im(g^*)$, we have for $h \in Hom(B,X)$,

$$f^*(h) = h \circ f = 0$$

Now, we can get the diagram

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "f"] & B \arrow[r, "g"] \arrow[d, "h"] & C \arrow[r] \arrow[dl, dashed] & 0 \\ & & X & & \end{tikzcd}$$

Notice that $Coker(f) = C$. Thus the morphism $C \to X$ is induced.

□

Similarly

♠ **Proposition 2.2.0.9** (Left Exactness of $Hom(X,-)$) Suppose $X \in \mathcal{A}$, then $Hom(-,X)$ is right-exact:For exact

$$0 \to A \to B \to C \to 0$$

Wwe can get a right-exact sequence

$$0 \to Hom(X,A) \to Hom(X,B) \to Hom(X,C)$$

∘ **Proof**

- $f^* : Hom(X,A) \to Hom(X,B)$ is injective: We have

$$f^*(h) = 0 \Rightarrow f \circ h = 0 \Rightarrow h = 0$$

- Similarly $Im(f^*) = Ker(g^*)$.

□

Furthermore, we have a longer exact sequence: Here

2.2.1 Homology Sequence

Definition 2.2.1.1 Define the homology sequence, with objects:

$$H_n(C) := Im(d_{n+1})/ker(d_n)$$

with the

♠ **Proposition 2.2.1.2** $H_\bullet(-)$ is a morphism in $Ch(-)$.

∘ **Proof** Omitted. □

Definition 2.2.1.3 Define: A morphism $C \to D$ is a **quasi-isomorphism** if the induced map

$$H_n(f) : H(C) \to H(D)$$

is an isomorphism.

2.2.2 Operations on Chain Complexes

Product, Direct Sum

Definition 2.2.2.1 For a family of chain complexes $\{C_\alpha\}$ in $Ch(\mathcal{A})$, where $\mathcal{A}$ is abelian category, define:

- Product $(\prod A_\alpha, \prod d_\alpha)$, with

$$\prod d_\alpha : \prod A_{\alpha,n} \to \prod A_{\alpha,n-1}$$

- Sum $(\oplus A_\alpha, \oplus d_\alpha)$, with

$$\oplus d_\alpha : \oplus A_{\alpha,n} \to \oplus A_{\alpha,n-1}$$

♠ **Proposition 2.2.2.2** The two operations commute with homology, i.e

$$\oplus H_n(A_\alpha) \cong H_n(\oplus A_\alpha) \qquad \prod H_n(A_\alpha) = H_n(\prod A_\alpha)$$

∘ **Proof** ... □

Subcomplex,Quotient Complex

Definition 2.2.2.3 A **subcomplex**

Especially: In Ab the quotient structure is defined as:

Definition 2.2.2.4 Given $A \hookrightarrow B$, then

$$A/B = coker(A \to B)$$

Thus, define the quotient complex

Definition 2.2.2.5 Define: For subcomplex $(B_n) \hookrightarrow (A_n)$ define

$$((A_n/B_n), (\bar{d}_n))$$

Induced by:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
\dots \arrow[r] & B_{n-1} \arrow[r] \arrow[d] & B_n \arrow[r] \arrow[d] & B_{n+1} \arrow[r] \arrow[d] & \dots \\
\dots \arrow[r] & A_{n-1} \arrow[r] \arrow[d] & A_n \arrow[r] \arrow[d] & A_{n+1} \arrow[r] \arrow[d] & \dots \\
\dots \arrow[r] & A_{n-1}/B_{n-1} \arrow[r] \arrow[d] & A_n/B_n \arrow[r] \arrow[d] & A_{n+1}/B_{n+1} \arrow[r] \arrow[d] & \dots \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

The exactness is given by 9-lemma.

Tensor Product

Definition 2.2.2.6 For $X, Y \in Ch_\bullet(A)$ write $X \otimes Y \in Ch_\bullet(A)$ for chain complex whose component is given by:

$$(X \otimes Y)_n := \oplus_{i+j=n} X_i \otimes_R Y_j$$

with $\partial^{x \otimes y}(x, y) = (\partial^X x, y) + (-1)^{deg(X)}(x, \partial^Y y)$.

♣ **Example 2.2.2.7** Let R be some ring and $I_\bullet \in C_\bullet(R - Mod)$ be the chain complex given by

$$I_\bullet = [\cdots \to 0 \to 0 \to R \overset{\partial_0^I}{\to} R \oplus R]$$

where $\partial_0^I = (-id, id)$. We may think of

$$I_0 = R \oplus R \cong R[\{(0), (1)\}] \qquad I_1 = R \cong R[(0 \to 1)]$$

Thus,

$$\partial_0^I : (0 \to 1) \mapsto (1) - (0)$$

Write out the tensor product $S_\bullet := I_\bullet \otimes I_\bullet$, by definition we have

$$S_0 =$$
$$S_1 =$$
$$S_2 =$$

♠ **Proposition 2.2.2.8** For category $\mathcal{A}$, we have

$$Ch(\mathcal{A}) \text{ is monoidal} \Leftrightarrow \mathcal{A} \text{ is monoidal}$$

∘ **Proof** ... □

Chain $[X, Y]$

Definition 2.2.2.9

- Define the category $Ch_\bullet^{ub}(\mathcal{A}) := \{C \in Ch_\bullet(\mathcal{A}) | C \text{ is unbounded}\}$
- Define:

$$[X, Y]_n := \prod_{i \in \mathbb{Z}} Hom_{RMod}(X_i, Y_{i+n})$$
$$df := d_Y \circ f - (-1)^n f \circ d_X \quad f \in [X, Y]_n$$

Thus,

$$[-, -] : Ch_\bullet(\mathcal{A})^{op} \times Ch_\bullet(\mathcal{A}) \to Ch_\bullet^{ub}(\mathcal{A})$$

is the **inner hom** of the category of chain complexes.

♠ **Proposition 2.2.2.10** The collection of cycles of the internal $Hom[X, Y]_\bullet$ in degree 0 coincides with the external hom functor

$$Z_0([X, Y]) = Hom_{Ch_\bullet^{ub}(X,Y)}$$

- **Proof** ... □

♠ **Proposition 2.2.2.11** The monoidal category $(Ch_\bullet, \otimes)$ is a closed monoidal category, the internal hom is the standard internal hom of chain complexes.

- **Proof** ... □

2.2.3 Shifting and Translation

Definition 2.2.3.1 Define the chain/cochain complex

$$C[p]_n := C_{n+p} \qquad C[p]^n = C^{n-p}$$

Then, define the differential map

$$d :=$$

2.2.4 Bicomplex

Definition 2.2.4.1 A **double complex** or **bicomplex** in $\mathcal{A}$ is a family $\{C_{p,q}\}$ of object $\mathcal{A}$ together with maps

$$d^h : C_{p,q} \to C_{p-1,q} \qquad d^v : C_{p,q} \to C_{p,q-1}$$

Such that: $d^h \circ d^h = d^v \circ d^v = d^v d^h + d^h d^v = 0$. Lattice

$$\begin{tikzcd}
& \cdots \arrow[d] & \cdots \arrow[d] & \cdots \arrow[d] & \\
\cdots \arrow[r] & C_{p-1,q+1} \arrow[r] \arrow[d] & C_{p-1,q} \arrow[r] \arrow[d] & C_{p-1,q-1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & C_{p,q+1} \arrow[r] \arrow[d] & C_{p,q} \arrow[r] \arrow[d] & C_{p,q-1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & C_{p+1,q+1} \arrow[r] \arrow[d] & C_{p+1,q} \arrow[r] \arrow[d] & C_{p+1,q-1} \arrow[r] \arrow[d] & \cdots \\
& \cdots & \cdots & \cdots &
\end{tikzcd}$$

Tot Complexes

Definition 2.2.4.2 Define:

$$Tot_n^{\prod} := \prod_{p+q=n} C_{p,q} \qquad Tot_n^{\oplus} = \oplus_{p+q=n} C_{p,q}$$

With the differential

$$d : Tot_n^{\prod} \to Tot_{n-1}^{\prod}$$
$$d : Tot_n^{\oplus} \to Tot_{n-1}^{\oplus}$$

such that $d \circ d = 0$.

♠ **Proposition 2.2.4.3** C is a bounded double complex with exact rows/columns$\Rightarrow$ $Tot(C)$ is acyclic.

∘ **Proof** ... □

2.2.5 Truncations,Translations

Truncations

Definition 2.2.5.1 If C is a chain complex, and n is an integer. We let $\tau_{\geq n}C$ denote the subcomplex of C defined by

$$(\tau_{\geq n}C)_i = \begin{cases} 0 & i < n \\ Z_n & i = n \\ C_i & i > n \end{cases}$$

Conversely, the quotient complex $\tau_{<n}C = C/(\tau_{\geq n}C)$

♠ **Proposition 2.2.5.2** Chearly that:

$$H_i(\tau_{\geq n}C) = \begin{cases} 0 & i < n \\ H_i(C) & i \geq n \end{cases}$$

∘ **Proof** ... □

Translations

Definition 2.2.5.3 Let C be a complex. Then, define

$$C[p]_n = C_{n+p} \qquad C[p]^n = C^{n-p}$$

More information will be given here:

2.3 Long Exact Sequence

2.3.1 9-Lemma

♠ **Proposition 2.3.1.1** (9-Lemma) Suppose the following commutative diagram

$$\begin{array}{ccccc} & 0 & 0 & 0 & \\ 0 & A' & B' & C' & 0 \\ 0 & A & B & C & 0 \\ 0 & A'' & B'' & C'' & 0 \\ & 0 & 0 & 0 & \end{array}$$

In abelian category $\mathcal{A}$, then:

- If the bottom two rows are exact, so is the top;
- If the top two rows are exact, so is the bottom row;
- If the top and bottom rows are exact, and the composite $A \to C$ is zero, the middle row is also exact;

∘ **Proof**

□

Snake's Lemma

★ **Corollary 2.3.1.2** (Snake's Lemma) Consider the following commutative diagram

$$\begin{tikzcd} & A \arrow[r] \arrow[d] & B \arrow[r] \arrow[d] & C \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & A' \arrow[r] & B' \arrow[r] & C' & \end{tikzcd}$$

There exists an exact sequence:

$$0 \longrightarrow ker(f) \longrightarrow ker(g) \longrightarrow ker(h) \longrightarrow Coker(f) \longrightarrow Coker(g) \longrightarrow Coker(h) \longrightarrow 0$$

∘ **Proof** ... □

5-Lemma

★ **Corollary 2.3.1.3** (5-Lemma) Consider the following diagram

$$\begin{tikzcd} 0 \arrow[r] & A_1 \arrow[r] \arrow[d] & A_2 \arrow[r] \arrow[d] & A_3 \arrow[r] \arrow[d] & A_4 \arrow[r] \arrow[d] & A_5 \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5 \arrow[r] & 0 \end{tikzcd}$$

with following properties

- If$\Rightarrow$
- If$\Rightarrow$
- If $\Rightarrow$

∘ **Proof** ... □

3-Lemma

★ **Corollary 2.3.1.4** (3-Lemma)For the diagram

$$\begin{matrix} 0 & A_1 & B_1 & C_1 & 0 \\ & & & & \\ 0 & A_2 & B_2 & C_2 & 0 \end{matrix}$$

◦ **Proof** ... □

Long Exact Sequence

▶ ***Theorem***

- Consider the following short exact sequence in $\mathcal{A}$:

$$0 \to A \to B \to C \to 0$$

  be an exact sequence, then we have a long exact sequence

$$\dots \to H_n(A) \to H_n(B) \to H_n(C) \xrightarrow{\delta} H_{n+1}(A) \to \dots$$

- Conversely, $H^{n+1}(A) \to H^n(C)$.

And in this sense, $H_\bullet$ is a covariant functor.

◦ **Proof**

-
-

□

2.3.2 Exact Triangle

Definition 2.3.2.1 ...

★ **Corollary 2.3.2.2** The category of chain complexes $Ch(\mathcal{A})$ is a **triangulated category**.

◦ **Proof** ... □

2.4 Chain Homotopies

Definition 2.4.0.1 A category

- has **enough projective** if for every obeject X if there is a projective object Q equipped with an epimorphism $Q \to X$;
- has **enough injectives** iff for every object X there is an injective Q equipped with monomorphism $X \to P$.

♠ **Proposition 2.4.0.2** Assuming the axiom of choice, the category $R - Mod$ has enough projective module.

◦ **Proof** ... □

2.4.1 Introduction

Consider the following vector space decomposition over field:

◇ **Lemma 2.4.1.1** Every exact sequence in $Vect_k$ splits.

$$C_n = Z_n \oplus B'_n; Z_n = Im(\partial_{n+1}), B'_n := C_n/Z_n = Coker(Z_n \to C_n) \cong B_{n-1}$$
$$Z_n = B_n \oplus H'_n, H'_n := Z_n/B_n = H_n(C)$$

$$\begin{tikzcd}
 & \cdots \arrow[d] & & & & \\
 & C_{n+1} \arrow[d, shift left, "\partial_{n+1}"] & 0 \arrow[d] & & & \\
0 \arrow[r] & Im(\partial_{n+1}) = B_n \arrow[u, dashed, shift left] \arrow[r, shift left] & Ker(\partial_n) = Z_n \arrow[l, dashed, shift left] \arrow[r] \arrow[d, shift left] & H_n(C_\bullet) \arrow[r] & 0 & \\
 & & C_n \arrow[u, dashed, shift left] \arrow[d, "\partial_n"] & & & \\
 & 0 \arrow[r] & Im(\partial_n) = B_{n-1} \arrow[r] \arrow[d] & Ker(\partial_{n-1}) = Z_{n-1} \arrow[r] \arrow[d] & H_{n-1}(C_\bullet) \arrow[r] & 0 \\
 & & 0 & C_{n-1} \arrow[d] & & \\
 & & & \cdots & &
\end{tikzcd}$$

Therefore, we can form thecompositions

$$C_n \xrightarrow{\pi} Z_n \xrightarrow{\pi} B_n \cong B'_{n+1} \subseteq C_{n+1}$$

By the subspace $C_{n+1}/Z_{n+1} \hookrightarrow C_{n+1}$. Define the **split map** $s_n : C_n \to C_{n+1}$. We have

$$\partial_{n+1} s_n \partial_{n+1} = \partial_{n+1}$$

Verify:

$$\cdots$$

And:

- ∂s:
- $s\partial$:

Split

Definition 2.4.1.2 A complex C is called split if there exists some $s_n : C_n \to C_{n+1}$ such that

$$d = dsd$$

such s is called **splitting map**.

Especially: If C is exact, then we say that C is **split exact**.

Chain Contruction

Definition 2.4.1.3

- We say that a chain map $f : C \to D$ is **null homotopic** if there are maps $s_n : C_n \to D_{n+1}$ such that

$$f = ds + sd$$

- The maps $\{s_n\}$ are called a **chain constraction** of f.

♠ **Proposition 2.4.1.4** For chain complex C_*, C_* splits exact⇔ id_{C_*} is null homotopic.

∘ **Proof**

- $\Rightarrow$ Then $id_{C_*} = ds + sd$. Now, we have

$$d = d(ds + sd) = dsd$$

- $\Leftarrow$ If id_{C_*} is null homotopic then there exists some s_n such that

$$id_{C_n} = d_{n+1}s_n + s_{n-1}d_n$$

(Exactness) Suppose that $a \in C_n$, $d_n(a) = 0, a \in Ker(d_n)$. Then,

$$\begin{aligned} &\Rightarrow s_{n-1}d_n(a) = 0 \\ &\Rightarrow a = d_{n+1}s_n(a) \\ &\Rightarrow a \in Im(d_{n+1}) \subseteq ker(d_n) \\ d^2 = 0 &\Rightarrow Im(d_{n+1}) = Ker(d_n) \end{aligned}$$

(Splitting)

$$d = d(ds + sd) = dsd$$

□

Chain Homotopies

Definition 2.4.1.5

- $f : C \to D$ is **null homotopic** if

$$f = ds + sd$$

- Consider two chain maps

$$f, g : C \to D$$

Then, f and g are **homotopic** iff $f - g$ is null homotopic. i.e.

$$f - g = sd + ds$$

The maps (s_n) are called **chain homotopy from** C to D.

- $f : C \to D$ is a **homotopy equivalence**

◇ **Lemma 2.4.1.6** (Chain Homotopy Invariance for $H_\bullet$)

-
-

∘ **Proof** ... □

♠ **Proposition 2.4.1.7**

- If $f : C \to D$ is null homotopic, then every map $f_* : H_n(C) \to H_n(D)$ is zero;
- If $f \simeq g$, then $H_n(f) = H_n(g) : H_n(C) \to H_n(D)$

∘ **Proof** ... □

Quotient Category

♠ **Proposition 2.4.1.8** Let K be a quotient category of Ch.Then, the functor

$$H_n : Ch(\mathcal{A}) \to \mathcal{A}$$

will factors through:

-
-
-
-

∘ **Proof** ... □

2.4.2 Resolution

2.5 Homology Exact Sequence

Definition 2.5.0.1 An **exact sqeunce** in $\mathcal{A}$ is a chain complex $C_\bullet$ in $\mathcal{A}$ with vanishing chain homology in each degree

$$\forall n \in \mathbb{N}, H_n(C) = 0$$

Especially, a **short exact sequence** is an exact sequence of the form

$$\cdots \to 0 \to A \to B \to C \to 0 \to \dots$$

Mostly written as

$$0 \to A \to B \to C \to 0$$

2.5.1 Some of Skills in Chain

3×3**-Lemma**

♠ **Proposition 2.5.1.1** ...

∘ **Proof** ... □

Snake Lemma

5-Lemma

Addenbum

2.5.2 Existence of ∂ in Exact Chain Complexes

So, the following diagram commutes:

$$H_*(A) \qquad\qquad\qquad H_*(B)$$

(The **exact triangle**)

$$H_*(C)$$

♠ **Proposition 2.5.2.1**

∘ **Proof** ... □

2.5.3 Resolution

♠ **Proposition 2.5.3.1** Suppose A_* be a resoltion of M, then

$$H_n(A_*) = \begin{cases} M & n = 0 \\ 0 & n \in \mathbb{Z} - \{0\} \end{cases}$$

∘ **Proof** Omitted. □

2.5.4 Functor Ext^1

Definition 2.5.4.1 Select an injective resolution of X:

$$0 \to X \to I^0 \to I^1 \to I^2 \to \dots$$

Apply $H(A, -)$ and deleting $Hom(A, X)$ we get

$$0 \to Hom(A, I^0) \xrightarrow{d^0} Hom(A, I^1) \xrightarrow{d^1} Hom(A, I^2) \to \dots$$

Define:

$$Ext^1(A, X) := ker(d^1)/im(d^0)$$

We have the following isomophic construction via projective resolution:

♠ **Proposition 2.5.4.2** In in the category of $\mathbb{Z}$-module/abelian groups,we have the exact sequence as followings:

$$\begin{tikzcd} 0 \arrow[r] & Hom(C,X) \arrow[r] & Hom(B,X) \arrow[r] & Hom(A,X) \arrow[dll, out=0, in=180] & \\ & Ext^1(C,X) \arrow[r] & Ext^1(B,X) \arrow[r] & Ext^1(A,X) \arrow[r] & 0 \end{tikzcd}$$

∘ **Proof** ... □

In General: In $\mathcal{A}b$-category we have the long exact sequence (here)

Extension and Ext

Ref: Weibel 3.4

Definition 2.5.4.3

- An **extension** of A by B is an exact sequence

$$0 \to A \to X \to B \to 0$$

- Two extensions of A by B are **equivalent** if

$$\begin{tikzcd} 0 \arrow[r] & B \arrow[r] \arrow[d, equal] & X \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & B \arrow[r] & X \arrow[r] & A \arrow[r] & 0 \end{tikzcd}$$

♣ **Example 2.5.4.4** (Extension of $\mathbb{Z}/p$) Consider the following equivalent class of split extensions:

$$0 \longrightarrow \mathbb{Z}/p \xrightarrow{p} \mathbb{Z}/p^2 \xrightarrow{i} \mathbb{Z}/p \longrightarrow 0 \text{ ,i=1,...,p-1}$$

◇ **Lemma 2.5.4.5** $Ext^1(A, B) = 0 \Leftrightarrow$ Every extension of A by B splits.

∘ **Proof** ... □

▶ ***Theorem*** ...

∘ **Proof** ... □

Definition 2.5.4.6 (Baer Sum)

2.6 Mapping Cones and Cylinders

2.6.1 Mapping Cones

Definition 2.6.1.1 Let $f : B \to C$ be a map of chain complexes. The **mapping cone** of f is the chain complex $cone(f)$ given by:

$$[cone(f)]_n := B_{n-1} \oplus C_n$$

With differentials

$$d(b, c) = (-d(b), d(c) - f(b)) \quad \begin{pmatrix} -d_B & 0 \\ -f & +d_C \end{pmatrix} : \quad \begin{tikzcd} B_{n-1} \arrow[r, "-"] \arrow[dr, "-"] & B_{n-2} \\ \oplus & \oplus \\ C_n \arrow[r, "+"] & C_{n-1} \end{tikzcd}$$

Then, we form a exact sequence of double complexes:

$$0 \xrightarrow{i} C \longrightarrow D = cone(f) \xrightarrow{p} B[-1] \longrightarrow 0$$

Especially $cone(C) := cone(id_C)$
Remark This is a double complex: Verify

$$\begin{tikzcd}
C_{n-1} \arrow[r] \arrow[d] & B_{n-2} \oplus C_{n-1} \arrow[r] \arrow[d] & B_{n-2} \arrow[d] \\
C_n \arrow[r] & B_{n-1} \oplus C_n \arrow[r] & B_{n-1}
\end{tikzcd}$$

We have $d(i(c)) = i(d(c)) = (0, d(c)); c \in C_{n-1}; p(d(b, c)) = d(p(b, c)) = d(c)$

And exactness by definition.

♠ **Proposition 2.6.1.2** $f : C \to D$ is **null homomtopic** iff if f extends $(-s, f) : cone(C) \to D$.

∘ **Proof** ... □

♠ **Proposition 2.6.1.3** With the exact sequence above, we have a long exact sequence

$$\cdots \to H_{n+1}(cone(f)) \to H_n(B) \to H_n(C) \to H_n(cone(f)) \to \dots$$

with $\partial = f_* : H_*(B[-1]) \to H_*(C)$

∘ **Proof** ... □

★ **Corollary 2.6.1.4** $f : B \to C$ is a quasi-isomorphism⇔ The mapping cone of $cone(f)$ is exact.

∘ **Proof** ... □

2.6.2 Mapping Cylinder

Definition 2.6.2.1 Define: For $f : B \to C$, the **mapping cylinder**

$$\begin{cases} cyl(f) := B_n \oplus B_{n-1} \oplus C_n \\ d(b, b', c) := (d(b) + b', -d(b'), d(c) - f(b')) \end{cases}$$

$$\begin{pmatrix} d_B & id_B & 0 \\ 0 & -d_B & 0 \\ 0 & -f & d_C \end{pmatrix} \Rightarrow \begin{pmatrix} d_B & id_B & 0 \\ 0 & -d_B & 0 \\ 0 & -f & d_C \end{pmatrix}^2 = \begin{pmatrix} d_B^2 & d_B - d_B & 0 \\ 0 & d_B^2 & 0 \\ 0 & f d_B - d_C f & d_C^2 \end{pmatrix} = 0$$

The diagram as following:

$$\begin{tikzcd}
B_n \arrow[r, "+"] & B_{n-1} \\
\oplus & \oplus \\
B_{n-1} \arrow[r, "-"] \arrow[uur, "+"] \arrow[ddr, "-"] & B_{n-2} \\
\oplus & \oplus \\
C_n \arrow[r, "+"] & C_{n-1}
\end{tikzcd}$$

◇ **Lemma 2.6.2.2** Let $cyl(C) := cyl(id_C)$. Then, it has

$$C_n \oplus C_{n-1} \oplus C_n$$

in degree n. Then, the two chain maps $f, g : C \to D$ are chain homotopic⇔ They extends to a map

$$(f, s, g) : cyl(C) \to D$$

∘ **Proof**

- $\Rightarrow$ If $f - g = sd + ds$ then we have

$$(f, s, g) : cyl(C) \to D \qquad (a, b, c) \mapsto f_n(a) + s_{n-1}(b) + g_n(c)$$

Then, we have

$$(f, s, g) \circ d_{Cyl(C)} = \begin{pmatrix} f & s & g \end{pmatrix} \begin{pmatrix} d & 1 & 0 \\ 0 & -d & 0 \\ 0 & -1 & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = fd + f - ds - g + gd = (s + f + g)d = d(s + f + g)$$

- $\Leftarrow$ Suppose that (f, s, g) is a chain map, then

$$(f, s, g) \circ d_{Cyl(C)} = fd + f - ds - g + gd = fd + gd + sd \Rightarrow f - g = ds + ds$$

□

◇ **Lemma 2.6.2.3** The subcomplex of elements $(0, 0, c)$ is isomorphic to C.
The corresponding inclusion

$$\alpha : C \to cyl(f)$$

is a quasi-isomorphism.

- **Proof** Firstly, this is a subcomplex. We have

$$d(0, 0, c) = (0, 0, d(c))$$

Then, $\alpha : C \to Cyl(f), c \mapsto (0, 0, c)$, then

$$H_n(\alpha) : H_n(C) \to H_n(cyl(f))$$

is a quasi-isomorphism: The quotient $cyl(f)/\alpha(C) = cone(-id_B)$.

...

So, it is null-homotopic, thus follows the long exact homology sequence

$$0 \to C \xrightarrow{\alpha} cyl(f) \to cone(-id_B) \to 0$$

□

♠ **Proposition 2.6.2.4** As above. $\beta : (b, b', c) \mapsto f(b) + c, cyl(f) \to C$ defines a chain map such that

$$\beta\alpha = id_C$$

And, $s(b, b', c) \to (0, b, 0)$ defines a chain homotopy from the identity of $cyl(f)$ to $\alpha\beta$.
α is in fact a chain homotopy equivalence between C and $cyl(f)$.

- **Proof** Consider the composition

...

□

Exact Sequence between

2.7 Euler Characteristics

2.7.1 Betti Numbers

2.7.2 Euler Characteristics

On Complex of Vector Spaces

Definition 2.7.2.1 Let R be a commutative ring, V be a chain complex of R-modules:

If each V_n is **finitely generated and projective**. Then, the **Euler characteristic** of V is the alternating sum of their ranks, if this is finite

$$\chi(V) = \sum (-1)^n rank_R(V_n)$$

On the Chain Complexes

▶ ***Theorem*** (Euler-Poincare)

$$\chi(C_\bullet) = \sum (-1)^i dim(H_i(C))$$

∘ **Proof** ... □

2.7.3 Euler Characteristic of a Topological Space

2.7.4 From Euler Characteristics to K_0 Groups

Chapter 3 Derived Functors

3.1 δ-Functors

Definition 3.1.0.1 A (covariant) homological/cohomological δ-functor between $\mathcal{A}$ and $\mathcal{B}$ is a collection of additive functors $T_n : \mathcal{A} \to \mathcal{B}$ and

$$\delta_n : T_n(C) \to T_{n-1}(A)$$
$$\delta^n : T^n(C) \to T^{n+1}(A)$$

defined for each short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$. Here we make the convention that $T^n = T_n = 0, n < 0$.

The two conditions are imposed:

- Long exact sequence

$$\cdots \to T_{n+1}(C) \xrightarrow{\delta} T_n(A) \to T_n(B) \to T_n(C) \xrightarrow{\delta} T_{n-1}(A) \to \dots$$
$$\cdots \to T^{n-1}(C) \xrightarrow{\delta} T^n(A) \to T^n(B) \to T^n(C) \xrightarrow{\delta} T^{n+1}(A) \to \dots$$

- For two exact $0 \to A \to B \to C \to 0$ and $0 \to A' \to B' \to C' \to 0$, we have commutes squares

$$\begin{tikzcd} T_n(C') \arrow[r, "\delta"] \arrow[d] & T_{n-1}(A') \arrow[d] \\ T_n(C) \arrow[r, "\delta"] & T_{n-1}(A) \end{tikzcd}$$

♣ **Example 3.1.0.2** Consider the functors

$$H_* : Ch_{\geq 0}(\mathcal{A}) \to \mathcal{A}$$
$$H^* : Ch^{\geq 0}(\mathcal{A}) \to \mathcal{A}$$

H_* gives a homological δ-functor, and H^* gives a cohomological δ-functor. See here:

♠ **Proposition 3.1.0.3** Let $\mathcal{S}$ be a category of short exact sequences

$$0 \to A \to B \to C \to 0 \qquad (*)$$

in $\mathcal{A}$. Then, δ_i is a natural transformation form the functor sending from the functor sending $(*)$ to $T_i(C)$, to the functor sending $(*)$ to $T_{i-1}(A)$.

- **Proof** Consider the two functors on on the category of short exact sequences $\mathcal{S}$ to $\mathcal{B}$:

$$\mathcal{F} : (0 \to A \to B \to C \to 0) \mapsto T_i(C)$$
$$\mathcal{G} : (0 \to A \to B \to C \to 0) \mapsto T_{i-1}(A)$$

Thus, the natural transformation

$$\begin{tikzcd} \mathcal{S} \arrow[r, bend left, "\mathcal{F}"] \arrow[r, bend right, "\mathcal{G}"'] \arrow[r, phantom, "\Downarrow"] & \mathcal{B} \end{tikzcd}$$

We could consider the following example:

$$\begin{array}{ccc} (0 \to A \to B \to C \to 0) & T_i(C) \longrightarrow T_{i-1}(A) \\ \downarrow{\scriptstyle (f,g,h)} & \downarrow{\scriptstyle T_i(h)} \qquad \downarrow{\scriptstyle T_{i-1}(f)} \\ (0 \to A' \to B' \to C' \to 0) & T_i(C') \longrightarrow T_{i-1}(A') \end{array}$$

□

♣ **Example 3.1.0.4** (p-torsion) If p is an integer,the functors $T_0(A) := A/pA$ and

$$T_1(A) = {}_pA = \{a \in A | pa = 0\}$$

fit together to form a homological δ-functor, or a cohomological δ-functor from Ab to Ab: Apply the snake lemma to

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r] \arrow[d, "p"] & B \arrow[r] \arrow[d, "p"] & C \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & A \arrow[r] & B \arrow[r] & C \arrow[r] & 0 \end{tikzcd}$$

We get the exact sequence:

$$0 \to {}_pA \to {}_pB \to {}_pC \xrightarrow{\delta} A/pA \to B/pB \to C/pC \to 0$$

♣ **Example 3.1.0.5** (Generalization) Similarly, for a ring R with element $r \in R$, then

$$T_0(M) := M/rM \qquad T_1 := {}_rM$$

And then, we will build a homological/cohomological functor $\delta : R - Mod \to \mathcal{A}b$,

$$T_n(M) = Tor_n^R(R/r, M)$$

Especially, if r is nonzerodivisor, equivalently to say is

$${}_rR := \{s \in R | rs = 0\} = 0$$

Then, we have

$$T_1^R(M) = Tor_1^R(R/r, M) = {}_rM; Tor_n(R/r, M) = 0, n \geq 2$$

And:

$$Tor_1^R(R/r, M) = {}_rM/\{sm | rs = 0, r \in R, m \in M\}$$

In general, ${}_rR \neq 0$.

3.1.1 Morphism of δ-Functors

Definition 3.1.1.1 Suppose S, T be two collection of δ-funtors. A **morphism of δ-functors** is a family of natural transformations between these functors, which commute with δ. i.e. For $0 \to A \to B \to C \to 0$ we have

$$\begin{tikzcd} S_n(C) \arrow[r, "\delta_n"] \arrow[d, "f_n(C)"] & S_{n-1}(A) \arrow[d, "f_{n-1}(A)"] \\ T_n(C) \arrow[r, "\delta_n"] & T_{n-1}(A) \end{tikzcd}$$

And

Definition 3.1.1.2

- A homological functor T is **universal**, if: Given
  - Another δ-Functor S;
  - A morphism $f_0 : S_0 \to T_0$

  There exists a unique morphism $\{f_n : S_n \to T_n\}$ of δ functor extends f_0;
- Dual Definition **couniversal** for extension $f_0 : T_0 \to S_0$ to $\{f_n : T_n \to S_n\}$.

♠ **Proposition 3.1.1.3** Functor

- $H_*(-) : Ch_{\geq 0}(\mathcal{A}) \to \mathcal{A}$ is universal δ-functor;
- $H^*(-) : Ch^{\geq 0}(\mathcal{A}) \to \mathcal{A}$ is universal δ-functor;

∘ **Proof** ... □

♠ **Proposition 3.1.1.4** Consider the following functor: Suppose $F : \mathcal{A} \to \mathcal{B}$ is exact, then the following functor:

$$T_0 = F \qquad T_n = 0, n \neq 0$$

Then, $T = \{T_n\}$ is a universal δ-functor.

∘ **Proof** ... □

3.2 Left Derived Functors

Definition 3.2.0.1 Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor between two abelian categories.

If $\mathcal{A}$ has enough projectives, we can construct the **left derived functors** $L_i F, i \geq 0$ of F as follows: If $A \in \mathcal{A}$, choose a projective resolution $P \to A$ and define

$$L_i F(A) := H_i(F(P))$$

Notice

$$\begin{tikzcd} \dots \arrow[r] & P_2 \arrow[r] \arrow[d] & P_1 \arrow[r] \arrow[d] & P_0 \arrow[d] \\ \dots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & A \end{tikzcd} \Rightarrow \text{Exact } P_1 \to P_0 \to A \Rightarrow \text{Exact } F(P_1) \to F(P_0) \to F(A) \to 0$$

Consider the exact sequence $F(P_1) \to F(P_0) \to F(A) \to 0$, we have

$$L_0 F(A) = H_0(F(P)) = \frac{ker(F(A) \to 0)}{Im(F(P_0) \to F(P_A))} = \frac{F(P_0)}{ker(F(P_0) \to F(A))} = Im(F(P_0) \to F(A)) = F(A)$$

3.2.1 Universal δ-Functority for $L_* F(-)$

The aim is to show $L_* F$ form a universal homological δ-functor.

◇ **Lemma 3.2.1.1** The objects $L_i F(A)$ of $\mathcal{B}$ are well defined up to isomorphism. That is: If $Q \to A$ is a second projective resolution, there exists a canonical isomorphism

$$L_i F(A) = H_i(F(P)) \xrightarrow{\cong} H_i(F(Q))$$

In particular, a different choice of projective resolution would yield new functor $\hat{L}_i F$, which are naturally isomorphic to the functors $L_i F$. This implies, that the value of $L_i F(A)$

- **Proof** Apply comparision theorem:

...

□

★ **Corollary 3.2.1.2** If A is projective, then $L_i F(A) = 0, i \neq 0$.

- **Proof** Consider the projective resolution of $0 \to P \xrightarrow{id} P \to 0$, then$L_i F(A) = H_i(F(P)) = 0$ □

3.2.2 F-Acyclic Functor

Definition 3.2.2.1 An object Q is called F**-Acyclic** if

$$L_i F(Q) = 0, i \neq 0$$

3.2.3 $L_i F$ is Universal δ-Functor

◇ **Lemma 3.2.3.1** If $f : A' \to A$ is any map in $\mathcal{A}$, there exists a natural map $L_i F(f) : L_i F(A') \to L_i F(A)$.

- **Proof** ... □

♠ **Proposition 3.2.3.2** ...

- **Proof** ... □

▶ ***Theorem*** Each $L_i F$ is an additive functor from $\mathcal{A}$ to $\mathcal{B}$

- **Proof** ... □

♠ **Proposition 3.2.3.3** (Preversing Derived Functor)

- **Proof** ... □

▶ ***Theorem*** The derived funcotr $L_* F$ form a homological δ-functor.

- **Proof** ... □

Dimension Shifting

♠ **Proposition 3.2.3.4** ...

∘ **Proof** ... □

Homology,Cohomology

♠ **Proposition 3.2.3.5** The functors $H_* : Ch_{\geq 0}(\mathcal{A}) \to \mathcal{A}$ and $H^* : Ch^{\geq 0}(\mathcal{A}) \to \mathcal{A}$ are universal δ-functors.

∘ **Proof** ... □

Effaceable Functors

Definition 3.2.3.6

- An additive funcotr $F : \mathcal{A} \to \mathcal{B}$ is **effaceable** if for each object A of $\mathcal{A}$ there is a monomorphism $u : A \to I$ such that $F(u) = 0$;
- Conversely, **coeffaceable** iff there exists surjection $u : P \to A$ such that $F(u) = 0$.

♠ **Proposition 3.2.3.7**

- If T_* is a homological δ-functor such that T_n is coeffaceable except T_0, then T_* is universal;
- If T^* is a cohomological δ-funcotr such that T_n os effacable, then T^* is universal.

∘ **Proof** ... □

3.3 Right Derived Functors

3.3.1 Right Derived Functors

Definition 3.3.1.1

$$R^i F(A) := H^i(F(I))$$

♠ **Proposition 3.3.1.2** (Right-Derived Functor is Dual of Left-Derived Functor)

∘ **Proof** ... □

3.3.2 Ext Functor

Ext Functor on R-modules

Definition 3.3.2.1 Define:

$$Ext^i_R(A, B) := R^i Hom_R(A, -)(B)$$

♠ **Proposition 3.3.2.2** (Criteria for Injective Modules) The followings are equivalent:

- B is injective R-module;
- $Hom(-, B)$ is exact;
- $Ext^i_R(A, B)$ vanishes for all $i \neq 0$ and all A; B is $Hom_R(-, B)$ acyclic for all A.
- $Ext^1_R(A, B)$ vanishes for all A.

∘ **Proof**

-
-
-
-

□

♠ **Proposition 3.3.2.3** (Criteria for Projective Modules)The followings are equivalent:

- A is projective;
- $Hom_R(A, -)$ is exact;
- $Ext^i_R(A, B)$ vanishes for all $i \neq 0$ and all B;
- $Ext^1_R(A, B)$ vanishes for all B.

∘ **Proof** ... □

Ext Functor in Abelian Categories

Definition 3.3.2.4

-
-

◇ **Lemma 3.3.2.5** The two concepts above are equivalent in abelian category.

∘ **Proof** ... □

♣ **Example 3.3.2.6** ...

Ext Functor in Derived Category

Definition 3.3.2.7 Consider the derived category $D(\mathcal{A})$. Define

$$Ext^p(X, A) := Hom_{D(A)}(X, A[p])$$

3.4 Adjoint Functors and Left/Right Exactness

▶ ***Theorem*** Suppose $L : \mathcal{A} \to \mathcal{B}$ and $R : \mathcal{B} \to \mathcal{A}$ be adjoint pair. Then, there exists a natural isomorphism

$$Hom(L(A), B) \xrightarrow{\cong} Hom(A, R(B))$$

∘ **Proof** ... □

★ **Corollary 3.4.0.1**

- $\otimes_R B : mod - R \to Ab$ is left adjoint to $Hom_{Ab}(B, -)$, so $- \otimes_R B$ is right exact;
- More generally, if B is an $R - S$ bimodule, then $\otimes_R B$ takes $mod - R$ to $mod - S$ and is a left adjoint, so it is right exact.

∘ **Proof** ... □

♠ **Proposition 3.4.0.2** ...

∘ **Proof** ... □

3.4.1 Tor Functor

Definition 3.4.1.1 Define: For functor $T(A) = A \otimes_R B$, define the abelian group

$$Tor_n^R(A, B) := (L_n T)(A)$$

In particular, $Tor_0^R(A, B) = A \otimes_R B$.
And when A

3.4.2 Adjoint Limit and Colimits

3.5 Balancing Tor and Ext

Recall: Tensor Complex $(A \otimes B, d \otimes 1, (-1)^p \otimes d)$, take the notation by

$$P \otimes_R Q \qquad Tot^{\oplus}(P \otimes_R Q)$$

3.5.1 Balancing Tor

▶ ***Theorem*** For R-module A, B we have:

$$L_n(A \otimes_R -)(B) = L_n(\otimes_R B)(A) = Tor_n^R(A, B); n \in \mathbb{N}$$

○ **Proof** ... □

★ **Corollary 3.5.1.1** We have:

$$Tor_n^R(\oplus_{i\in I}A_i, B) \cong \oplus_{i\in I}Tor_n^R(A_i, B)$$
$$Tor_n^R(A, \oplus_{i\in I}B_i) \cong \oplus_{i\in I}Tor_n^R(A, B_i)$$

○ **Proof** ... □

A question is: When can we state:

$$Tor_n^R(\prod A_i, B) \cong \prod Tor_n^R(A_i, B)$$

- $n = 0$: $Tor_0^R(\prod A_i, B) \Leftrightarrow$
- $n \geq 1$: $\Leftrightarrow$ R is right-coherent ring, and B is finitely generated module.

3.5.2 Balancing Ext

▶ ***Theorem*** For R-module A, B we have

$$Ext_R^n(A, B) = R^n Hom_R(A, -)(B) = R^n Hom_R(-, B)(A)$$

○ **Proof** ... □

★ **Corollary 3.5.2.1** We have

$$Ext_R^n(A, \prod_{j\in J} B_j) = \prod_{j\in J} Ext_R^n(A, B_j)$$
$$Ext_R^n(\oplus_{i\in I}A_i, B) = \prod_{i\in I} Ext_R^n(A_i, B)$$

○ **Proof** ... □

More discussions in next chapter.

Chapter 4 Tor and Ext

4.1 Tor for Abelian Groups

♠ **Proposition 4.1.0.1** Suppose A, B are abelian groups. Then we have

$$Tor_n^{\mathbb{Z}}(A, B) = \begin{cases} \mathbb{Z} & n = 1 \\ 0 & n > 1 \end{cases}$$

∘ **Proof** ... □

▶ ***Theorem*** (Acyclic Assembly Lemma) Let C be a double complex in $mod - R$, then

-
-

∘ **Proof** ... □

Definition 4.1.0.2 (Hom-Cochain Complex)

▶ ***Theorem*** For every pair of R-modules A and B, and all n,

$$Ext_R^n(A, B) = R^n Hom_R(A, -)(B) = R^n Hom_R(-, B)(A)$$

∘ **Proof** ... □

Definition 4.1.0.3 ...

4.2 Tor and Flatness

Definition 4.2.0.1 Recall:

- Consider $B \in R - Mod$. Then, B is **flat** iff $- \otimes B$ is exact.
- Dual to define $A \in Mod - R$. Then, A is **flat** iff $A \otimes -$ is exact.

Pontrjagin Dual

Definition 4.2.0.2 The Pontrjagin dual B^* of a left-module B is the right R-module

$$B \in Hom_{Ab}(B, \mathbb{Q}/\mathbb{Z})$$

and element $r \in R$ acts via $(fr)(b) = f(rb)$. So B^* is a right R-module.

▶ ***Theorem*** For any $R - moudle$ we have

$$Ext^n_R(A, B^\vee) \cong (Tor^R_n(A, B))^\vee$$

∘ **Proof** ... □

4.3 Ext for Nice Rings

◇ **Lemma 4.3.0.1** Let A, B be abelian groups, then we have

$$Ext^n_{Ab}(A, B) = \begin{cases} \\ 0 & n > 1 \end{cases}$$

∘ **Proof** ... □

4.4 Ext and Extensions

4.5 Derived Functor and Inverse Limits

Notion In this section, we take

- $\mathcal{A} = \mathcal{A}b$;
- I is the poset $\cdots \to 2 \to 1 \to 0$;

Definition 4.5.0.1 (Tower of Abelian Groups) Define

$$\{A_i\} : \cdots \to A_2 \to A_1 \to A_0$$

4.5.1 Mittag-Leffers System

Definition 4.5.1.1 (Mittag-Leffers System) Let I be a directed set, (A_i, φ_{ij}) be an inverse system in $Sets$.

Limit

◇ **Lemma 4.5.1.2** (Existence)

∘ **Proof** ... □

◇ **Lemma 4.5.1.3** (Exactness) Let

...

4.6 Universal Coefficient Theorem

▶ ***Theorem*** (Künneth)

◦ **Proof** ... □

▶ ***Theorem*** (UCT for Homology)

◦ **Proof** ... □

▶ ***Theorem*** (UCT for Cohomology)

◦ **Proof** ... □

4.6.1 Topology

▶ ***Theorem*** (UCT in Topology)

◦ **Proof** ... □

Chapter 5 Homological Dimension

5.1 Dimension

Define:

Definition 5.1.0.1

- **The Projective Dimension**:
- **The Injective Dimension**:
- **The Flat Dimension**

Theorem (Global Dimension Theorem) The following numbers are the same for any ring R:

-
-
-
-

Proof ... □

Definition 5.1.0.2 ...

Noetherian Ring

5.1.1 Rings of Small Dimension

5.2 Local Rings

5.3 Konzul Complexes

Definition 5.3.0.1 (Konzul Complex) Take the following notion

$$...$$

5.4 Local Cohomology

5.5 Appendix: Hilbert-Serre Theorem

Our main reference:

- https://faculty.ustc.edu.cn/yqliang/files/myarticle/Serrethmonnoetherianregularlocalring.pdf

▶ ***Theorem*** (Serre)

◦ **Proof** ... □

Chapter 6 Spectral Sequence

6.1 Introductions

- (Leray) Sheaf, Spectral Sequence;
- (Koszul,1945) Algebraic;

6.1.1 Spectral Sequence

Definition 6.1.1.1 Let $\mathcal{A}$ be an abelian category

- A **spectral sequence** in $\mathcal{A}$ is given by a system

$$(E_r, d_r)_{r\geq 1}$$

where: Each E_r is an object of $\mathcal{A}$, each $d_r : E_r \to E_r$ is a morphism such that

$$d_r \circ d_r = 0 \qquad E_{r+1} = ker(d_r)/Im(d_r)$$

- A morphism of spectral sequences

$$f : (E_r, d_r)_{r\geq 1} \to (E'_r, d'_r)_{r\geq 1}$$

is given by a family of morphisms $(f_r)_{r\geq 1}$ such that

$$f_r \circ d_r = d'_r \circ f_r$$

And $f_{r+1} : E_{r+1} \to E'_{r+1}$ is indentified by the morphism induced by $f_r : E_r \to E'_r$

$$\begin{tikzcd}
Im(d_r) \arrow[r, "\subseteq"] \arrow[d, "f_r|Im(d_r)"] & ker(d_r) \arrow[r] \arrow[d, "f|ker(d_r)"] & E_{r+1} \arrow[d, dashed] \\
Im(d'_r) \arrow[r, "\subseteq"] & ker(d'_r) \arrow[r] & E'_{r+1}
\end{tikzcd}$$

Remark Sometimes denoted as $(E_r, d_r)_{r\geq 0}$.We loose the defition: There exists isomorphism $E_r \to ker(d_r)/Im(d_r)$.

Definition 6.1.1.2 ...

Spectral Sequence

Definition 6.1.1.3 Given a double complex $(E_r, d_r)_{r\geq 1}$, define

$$0 = B_1 \subseteq B_2 \subseteq \cdots \subseteq Z_r \subseteq \cdots \subseteq Z_2 \subseteq Z_1 = E_1$$

By following procedure:

$$B_{r+1} = Im(d_r) \qquad Z_{r+1} = Ker(d_r)$$

Then, it is clear that $d_2 : Z_2/B_2 \to Z_2/B_2$.

Limit of Spectral Sequence

Definition 6.1.1.4 Let $\mathcal{A}$ be an abelian category, $(E_r, d_r)_{r\geq 1}$ be a spectral sequence.

- If the subobjects $Z_\infty = \cap Z_r$ and $B_\infty = \cup B_r$ of E_1 exist. Then, we define the **limit of spectral sequence** to be the object

$$Z_\infty = \cap Z_r \qquad B_\infty = \cup B_r$$

- We say that the spectral sequence degenerates at E_r if the differentials $d_r, d_{r+1}, \dots$ are all zero.

If the spectral sequence degenerates at E_r,then we have

$$E_r = E_{r+1} = \cdots = E_\infty$$

Also, almost any abelian category we will encounter has countable sums and intersections.

6.1.2 Exact Couple

Definition 6.1.2.1 Let $\mathcal{A}$ be an exact category:

- An **exact couple** is a datum (A, E, α, f, g) where A, E are objects of $\mathcal{A}$ and α, f, g are morphisms as in following diagram

$$\begin{tikzcd} A \arrow[rr, "\alpha"] & & A \arrow[dl, "g"] \\ & E & \end{tikzcd}$$

-

◇ **Lemma 6.1.2.2** ...

∘ **Proof** ... □

6.1.3 Differential Object

Definition 6.1.3.1 Let $\mathcal{A}$ be an abeian category.
A **differential object** of $\mathcal{A}$ is a pair (A, d) consisting of

6.1.4 Filtered Differential Object

6.1.5 Filtered Complexes

6.1.6 Double Complexes

6.1.7 Abelian Groups

Chapter 7 Simplicial Methods in Homological Algebra

7.1 Simplicial Objects

7.2 Operations on Simplicial Objects

Chapter 8 Hochshild and Cyclic Homology

Chapter 9 Derived Category

9.1 Triangulated Category

Definition 9.1.0.1 Define: Let $\mathcal{D}$ be an additive category, let

$$[1] : \mathcal{D} \to \mathcal{D} \qquad E \mapsto E[1]$$

be an additive functor which is an auto-equivalence of $\mathcal{D}$:

- A **triangle** is sextuple

$$(X, Y, Z, f, g, h)$$

  where

  - $X, Y, Z \in Ob(\mathcal{D})$;
  - $f : X \to Y; g : Y \to Z; h : Z \to X[1]$ are morphisms of $\mathcal{D}$;

- A morphism of triangles

$$(X, Y, Z, f, g, h) \to (X', Y', Z', f', g', h')$$

  with morphisms

$$\begin{cases} a : X \to X' \\ b : Y \to Y' \\ c : Z \to Z' \end{cases}$$

  of $\mathcal{D}$ such that

$$\begin{cases} b \circ f = f' \circ a \\ c \circ g = g' \circ b \\ a[1] \circ h = h' \circ c \end{cases}$$

With visualization

$$\begin{array}{cccc} X & Y & Z & X[1] \\ \\ X' & Y' & Z' & X'[1] \end{array}$$

9.1.1 Triangulated Category

Definition 9.1.1.1 Axioms:

- TR1: Any triangle category isomorphic to a distinguished triangle is a distinguished triangle.
  Any triangle of the form $(X, X, 0, id, 0, 0)$ is distinguished.
  For any morphism $f : X \to Y$ in $\mathcal{D}$, there exists a **distinguished triangle** of the form

$$(X, Y, Z, f, g, h)$$

- TR2: The triangle (X, Y, Z, f, g, h) is distinguished iff $(Y, Z, X[1], g, h, -f[1])$ is
- TR3: Given the diagram

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

$$X' \longrightarrow Y' \longrightarrow Z' \longrightarrow X[1]$$

- TR4: Given objects $X, Y, Z \in Ob(\mathcal{D})$, and morphisms $f : X \to Y, g : Y \to Z$ and distinguished triple

$$(X, Y, Q_1, f, p_1, d_1) \quad (X, Z, g\circ f, p_2, d_2) \quad (Y, Z, Q_2, g, p_3, d_3)$$

  There exists morphisms

$$a : Q_1 \to Q_2 \qquad b : Q_2 \to Q_3$$

  such that:

  - $(Q_1, Q_2, Q_3,)$
  -
  -

We will call:

-
-

Remark

Homological Functor

Definition 9.1.1.2

- Let $\mathcal{D}$ be a pre-triangulated category, $\mathcal{A}$ be an abelian category.
  An additive category $H : \mathcal{D} \to \mathcal{A}$ is **homological** iff for every distinguished triangle (X, Y, Z, f, g, h) the sequence

$$H(X) \to H(Y) \to H(Z)$$

  is exact in the abelian category $\mathcal{A}$

- Conversely, $H : \mathcal{D}^{op} \to \mathcal{A}$ if the corresponding functor

$$H^{op} : \mathcal{D} \to \mathcal{A}^{op}$$

  is homological.

.

Remark If $H : \mathcal{D} \to \mathcal{A}$, we write

$$H^n(X) := H(X(n))$$

Definition 9.1.1.3 By TR2, a distinguished triangle (X, Y, Z, f, g, h) determines a long exact sequence

$$H^{-1}(Z) \overset{above}{\to} H^0(X) \overset{above}{\to} H^0(Y) \overset{above}{\to} H^0(Z) \overset{H(h)}{\to} H^1(X)$$

δ-Functor

Definition 9.1.1.4 Let $\mathcal{A}$ be an abelian category. Let $\mathcal{D}$ be a triangulated category.
A δ-functor from $\mathcal{A}$ to $\mathcal{D}$ is given by a functor

$$G : \mathcal{A} \to \mathcal{D}$$

9.1.2 Elementray Results

◇ **Lemma 9.1.2.1** Let $\mathcal{D}$ be a pre-triangulated category, let (X, Y, Z, f, g, h) be distinguished triangle, then

$$g \circ f = 0 \quad h \circ g = 0 \quad f[1] \circ h = 0$$

∘ **Proof** ... □

◇ **Lemma 9.1.2.2** Let $\mathcal{D}$ be a pre-triangulated category. For any object W of $\mathcal{D}$, the functor $Hom_{\mathcal{D}}(W, -)$ is homological, and the functor $Hom_{\mathcal{D}}(-, W)$ is cohomological.

∘ **Proof** ... □

◇ **Lemma 9.1.2.3** Let $\mathcal{D}$ be a pretriangulated category, and

$$(a, b, c) : (X, Y, Z, f, g, h) \to$$

∘ **Proof** ... □

♠ **Proposition 9.1.2.4** Let $\mathcal{D}$ be trianglatedd category. Any commutative category

$$\begin{matrix} X & Y \\ X' & Y' \end{matrix}$$

can be extended to the diagram:

$$\begin{matrix} X & Y & Z & X[1] \\ X' & Y' & Z' & X'[1] \\ X'' & Y'' & Z & X''[1] \\ X[1] & Y[1] & Z[1] & X[2] \end{matrix}$$

where: All square commutes, except for the lower right squrae which is anticommutative.

Moreover, each of column and rows are distinguished triangles.

Finally, the morphisms on the bottom row/right column are obtained from the morphisms of the top row/left column applying [1].

- **Proof** ... □

9.1.3 Constructions

Triangulated Functor

Localization of Triangulated Functor

Definition 9.1.3.1 Let: $\mathcal{D}$ be a pre-triangulated category, we say a multiplicative system S is **compatible with triangulated stucture** if the following two conditions hold:

- MS5: For a morphism f of $\mathcal{D}$ we have

$$f \in S \Leftrightarrow f[1] \in S$$

- Given a solid commutative square

$$\begin{tikzcd} X \arrow[r] \arrow[d, "s"] & Y \arrow[r] \arrow[d, "s'"] & Z \arrow[r] \arrow[d, dashed, "\exists s''"] & X[1] \arrow[d, "s[1]"] \\ X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & X'[1] \end{tikzcd}$$

whose rows are distinguished triangles with $s, s' \in S$, there exists a morphism

$$s'' : Z \to Z'$$

in S such that (s, s', s'') is a morphism of triangles.

◇ **Lemma 9.1.3.2** ...

Quotient of Triangulated Category

◇ **Lemma 9.1.3.3**

9.2 The Homotopy Category

Definition 9.2.0.1 Let $\mathcal{A}$ be additive category:

- We set $Comp(\mathcal{A}) = CoCh(\mathcal{A})$ be the **category of (cochain) complexes**.
- A complex $K^\bullet$ is **bounded below** if:

...

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-
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9.3 Derived Category

9.3.1 Filtered Derived Category

Chapter 10 Appendix

10.1 Eckmann-Hilton

Theorem Suppose X be a structure, with two binary operations $\circ, \star$ which satisfies

- (Identity) $x \circ e = e \circ x = x; x \star e = e \star x = x$;
- (Eckmann-Hilton) For $a, b, c, d \in X$,

$$(a \circ b) \star (c \circ d) = (a \star c) \circ (b \star d)$$

  Then:

  - $a \circ b = a \star b$;
  - The operation is commutative;
  - The operation is associative.

Proof Notice

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$\square$
